

DUBAI RAPID LINK CONSORTIUM

**Dubai Metro, UAE
Jebel Ali Depot**

**Supplementary Design Calculation
Building 8 – DCC / Administration Building**

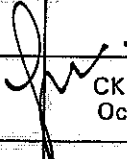
October 2007

ATKINS

WS Atkins & Partners Overseas
Supported by Meinhardt Infrastructure

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1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (P. Shepherd-Smith)
4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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Rev A3

Date: 26 Oct 2007

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1 Introduction

INTRODUCTION

This calculation report is submitted as a supplementary calculation to reply engineer's comment to our DCP3 submission via transmittal no. 007670 dated 20 Jun 2007.

Please refer to previously submitted Document no. DM001-E-AAL-CDP-DR-DCC-373902 Rev A1 & A2 for the design calculation of all other structural members.

2 Footing Design

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site: Dubai LRT	Date: 2/8/2007	Job No. 068010.01
Footing F7	Designed by LM	Sheet No.
For soil bearing pressure = 450kPa		

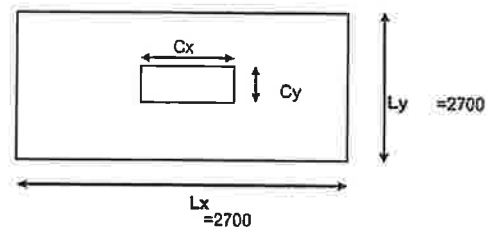
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	2700	mm
Length of Footing	L_x	=	2700	mm
Width of Column	C_y	=	600	mm
Length of Column	C_x	=	600	mm
Concrete Ten. Cover	c	=	75	mm
Concrete Comp. Cover	c	=	75	mm
Tension bar dia. (x-dir)	D_x	=	20	mm
Tension bar dia. (y-dir)	D_y	=	20	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	515	mm
Effect.depth (y-dir)	d_y	=	495	mm
Area of footing	A	=	7.29	m ²



3. Loadings

Safe bearing pressure	p_b	=	450	kPa
Load factor	f	=	1.5	
Design axial load (SLS)	N	=	3040.00	kN
Soil bearing pressure		=	417.01	kPa
Eccentricity	e	=	0	mm

< 450 kPa ok

4. Analysis

Net pressure (SLS)	p_{max}	=	435.60	kPa			
Net pressure (ULS)	p_{max}	=	653.4	kPa			
Shear force (x-dir)	V_x	=	1852.39	kN			
Shear force (y-dir)	V_y	=	1852.39	kN			
Moment (x-dir)	M_x	=	972.50	kNm			
Moment (y-dir)	M_y	=	972.50	kNm			
Steel required (x-dir)	A_{sreq}	=	4967	mm ²			
Steel provided (x-dir)	A_{sprov}	=	5655	mm ²	O.K.	Nos	18
Steel required (y-dir)	A_{sreq}	=	5168	mm ²			
Steel provided (y-dir)	A_{sprov}	=	5655	mm ²	O.K.	Nos	18
Top bars required (x-dir)	A'_{sreq}	=	2106	mm ²			
Top bars provided (x-dir)	A'_{sprov}	=	3619	mm ²	O.K.	Nos	18
Top bars required (y-dir)	A'_{sreq}	=	2106	mm ²			
Top bars provided (y-dir)	A'_{sprov}	=	3619	mm ²	O.K.	Nos	18

Spacing			
149	x-dir		
149	y-dir		
149	x-dir		
149	y-dir		

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	2/8/2007	Job No.	068010.01
	Footing F7				
	For soil bearing pressure = 450kPa	Designed by	LM	Sheet No.	

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 4677.75$ kN
Shear stress at face $v_f = 3.78$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	515	3127.34	0.93	0.55	1.10	No Shear Links
2d	1030	144.72	0.03	0.55	0.55	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 1913.63$ kN
Shear stress at face (x-dir) $v_c = 1.38$ N/mm² O.K.

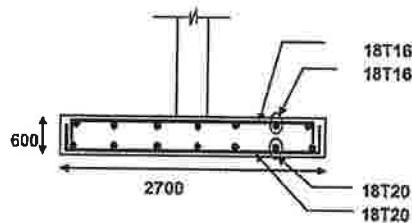
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	515	975.04	0.70	0.55	1.10	No Shear Links
2d	1030	36.45	0.03	0.55	0.55	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 1913.63$ kN
Shear stress at face (y-dir) $v_c = 1.43$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	495	1011.49	0.76	0.55	1.11	No Shear Links
2d	990	109.35	0.08	0.55	0.55	No Shear Links

6. Detailing



Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site: Dubai LRT	Date: 2/8/07	Job No. 068010.01
Footing F24	Designed by LM	Sheet No.
For soil bearing pressure = 450kPa		

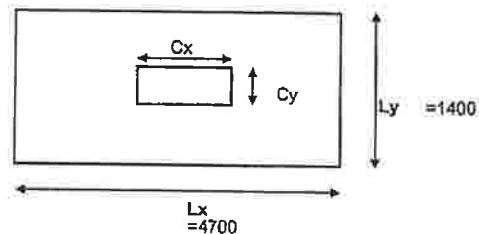
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	750	mm
Width of Footing	L_y	=	1400	mm
Length of Footing	L_x	=	4700	mm
Width of Column	C_y	=	300	mm
Length of Column	C_x	=	4000	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	692	mm
Effect.depth (y-dir)	d_y	=	676	mm
Area of footing	A	=	6.58	m ²



3. Loadings

Safe bearing pressure	p_b	=	450	kPa
Load factor	f	=	1.5	
Design axial load (SLS)	N	=	2936.00	kN
Soil bearing pressure		=	446.20	kPa
Eccentricity	e	=	0	mm

< 450 kPa ok

4. Analysis

Net pressure (SLS)	p_{max}	=	432.00	kPa
Net pressure (ULS)	p_{max}	=	648	kPa
Shear force (x-dir)	V_x	=	317.52	kN
Shear force (y-dir)	V_y	=	1675.08	kN
Moment (x-dir)	M_x	=	55.57	kNm
Moment (y-dir)	M_y	=	460.65	kNm
Steel required (x-dir)	A_{sreq}	=	1365	mm ²
Steel provided (x-dir)	A_{sprov}	=	1608	mm ²
Steel required (y-dir)	A_{sreq}	=	4583	mm ²
Steel provided (y-dir)	A_{sprov}	=	5027	mm ²
Top bars required (x-dir)	A'_{sreq}	=	1365	mm ²
Top bars provided (x-dir)	A'_{sprov}	=	1608	mm ²
Top bars required (y-dir)	A'_{sreq}	=	4583	mm ²
Top bars provided (y-dir)	A'_{sprov}	=	5027	mm ²

	Nos	ϕ	Spacing	
O.K.	8	16	183	x-dir
O.K.	25	16	191	y-dir
O.K.	8	16	183	x-dir
O.K.	25	16	191	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	2/8/07	Job No.	068010.01
	Footing F24				
	For soil bearing pressure = 450kPa	Designed by	LM	Sheet No.	

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 3631.50$ kN
Shear stress at face $v_f = 0.61$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_e (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	692	0.00	0.00	0.41	0.81	No Shear Links
2d	1384	0.00	0.00	0.41	0.41	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_e = 330.75$ kN
Shear stress at face (x-dir) $v_e = 0.34$ N/mm² O.K.

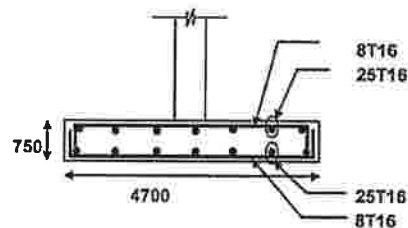
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_e (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	692	0.00	0.00	0.41	0.81	No Shear Links
2d	1384	0.00	0.00	0.41	0.41	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_e = 1744.88$ kN
Shear stress at face (y-dir) $v_e = 0.55$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_e (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	676	0.00	0.00	0.40	0.80	No Shear Links
2d	1352	0.00	0.00	0.40	0.40	No Shear Links

6. Detailing



Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT Footing F10a For soil bearing pressure = 450kPa	Date	2/8/2007	Job No.	068010.01
		Designed by	LM	Sheet No.	

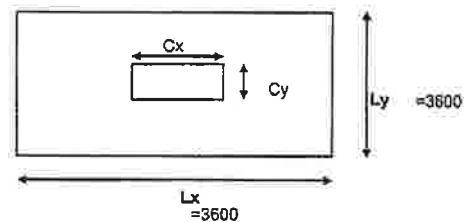
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	900	mm
Width of Footing	L_y	=	3600	mm
Length of Footing	L_x	=	3600	mm
Width of Column	C_y	=	600	mm
Length of Column	C_x	=	600	mm
Concrete Ten. Cover	c	=	75	mm
Concrete Comp. Cover	c	=	75	mm
Tension bar dia. (x-dir)	D_x	=	25	mm
Tension bar dia. (y-dir)	D_y	=	25	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	812.5	mm
Effect.depth (y-dir)	d_y	=	787.5	mm
Area of footing	A	=	12.96	m ²



3. Loadings

Safe bearing pressure	P_b	=	450	kPa
Load factor	f	=	1.5	
Design axial load (SLS)	N	=	5598.00	kN
Soil bearing pressure		=	431.94	kPa
Eccentricity	e	=	0	mm

< 450 kPa ok

4. Analysis

Net pressure (SLS)	p_{max}	=	428.40	kPa
Net pressure (ULS)	p_{max}	=	642.6	kPa
Shear force (x-dir)	V_x	=	3470.04	kN
Shear force (y-dir)	V_y	=	3470.04	kN
Moment (x-dir)	M_x	=	2602.53	kNm
Moment (y-dir)	M_y	=	2602.53	kNm
Steel required (x-dir)	A_{sreq}	=	8425	mm ²
Steel provided (x-dir)	A_{sprov}	=	12272	mm ²
Steel required (y-dir)	A_{sreq}	=	8692	mm ²
Steel provided (y-dir)	A_{sprov}	=	12272	mm ²
Top bars required (x-dir)	A'_{sreq}	=	4212	mm ²
Top bars provided (x-dir)	A'_{sprov}	=	5027	mm ²
Top bars required (y-dir)	A'_{sreq}	=	4212	mm ²
Top bars provided (y-dir)	A'_{sprov}	=	5027	mm ²

O.K.	Nos	ϕ	Spacing	
	25	25	143	x-dir
O.K.	Nos	ϕ	Spacing	
	25	25	143	y-dir
O.K.	Nos	ϕ	Spacing	
	25	16	143	x-dir
O.K.	Nos	ϕ	Spacing	
	25	16	143	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	2/8/2007	Job No.	088010.01
	Footing F10a	Designed by	LM	Sheet No.	
	For soil bearing pressure = 450kPa				

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 8505.00$ kN
Shear stress at face $v_f = 4.36$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	812.5	5406.33	0.75	0.55	1.11	No Shear Links
2d	1625	0.00	0.00	0.55	0.55	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_x = 3645.00$ kN
Shear stress at face (x-dir) $v_x = 1.25$ N/mm² O.K.

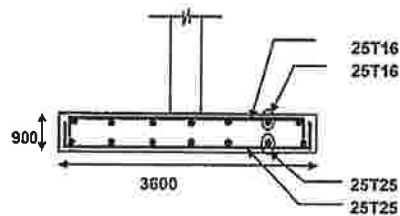
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	812.5	1670.63	0.57	0.55	1.11	No Shear Links
2d	1625	0.00	0.00	0.55	0.55	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_y = 3645.00$ kN
Shear stress at face (y-dir) $v_y = 1.29$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_e (N/mm ²)	Remarks
d	787.5	1731.38	0.61	0.56	1.12	No Shear Links
2d	1575	0.00	0.00	0.56	0.56	No Shear Links

6. Detailing



Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	2/8/07	Job No.	088010.01
	Footing F26				
	For soil bearing pressure = 450kPa	Designed by	LM	Sheet No.	

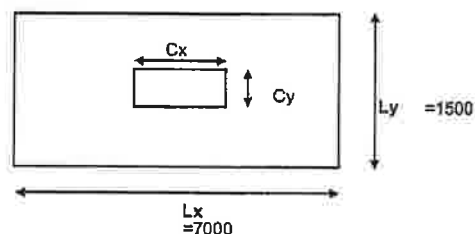
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	750	mm
Width of Footing	L_y	=	1500	mm
Length of Footing	L_x	=	7000	mm
Width of Column	C_y	=	300	mm
Length of Column	C_x	=	6600	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_o	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect depth (x-dir)	d_x	=	692	mm
Effect depth (y-dir)	d_y	=	676	mm
Area of footing	A	=	10.5	m ²



3. Loadings

Safe bearing pressure	p_b	=	450	kPa
Load factor	f	=	1.5	
Design axial load (SLS)	N	=	4701.00	kN
Soil bearing pressure		=	447.71	kPa
Eccentricity	e	=	0	mm

< 450 kPa ok

4. Analysis

Net pressure (SLS)	p_{max}	=	432.00	kPa
Net pressure (ULS)	p_{max}	=	648	kPa
Shear force (x-dir)	V_x	=	194.40	kN
Shear force (y-dir)	V_y	=	2721.60	kN
Moment (x-dir)	M_x	=	19.44	kNm
Moment (y-dir)	M_y	=	816.48	kNm
Steel required (x-dir)	$A_{s,req}$	=	1463	mm ²
Steel provided (x-dir)	$A_{s,prov}$	=	1810	mm ²
Steel required (y-dir)	$A_{s,req}$	=	6825	mm ²
Steel provided (y-dir)	$A_{s,prov}$	=	7037	mm ²
Top bars required (x-dir)	$A'_{s,req}$	=	1463	mm ²
Top bars provided (x-dir)	$A'_{s,prov}$	=	1810	mm ²
Top bars required (y-dir)	$A'_{s,req}$	=	6825	mm ²
Top bars provided (y-dir)	$A'_{s,prov}$	=	7037	mm ²

	Nos	ϕ	Spacing	
O.K.	9	16	173	x-dir
O.K.	35	16	202	y-dir
O.K.	9	16	173	x-dir
O.K.	35	16	202	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai LRT	Date	2/8/07	Job No.	068010.01
	Footing F26				
	For soil bearing pressure = 450kPa	Designed by	LM	Sheet No.	

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 5751.00$ kN
Shear stress at face $v_f = 0.60$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	692	0.00	0.00	0.41	0.83	No Shear Links
2d	1384	0.00	0.00	0.41	0.41	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_x = 202.50$ kN
Shear stress at face (x-dir) $v_x = 0.20$ N/mm² O.K.

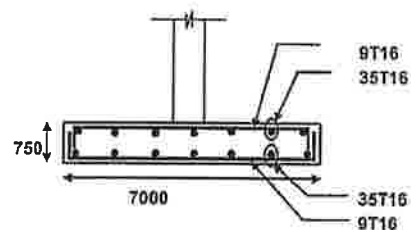
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	692	0.00	0.00	0.41	0.83	No Shear Links
2d	1384	0.00	0.00	0.41	0.41	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_y = 2835.00$ kN
Shear stress at face (y-dir) $v_y = 0.60$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	676	0.00	0.00	0.39	0.78	No Shear Links
2d	1352	0.00	0.00	0.39	0.39	No Shear Links

6. Detailing



3 Beam Design

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site *Arba' Netno.*

Date

Job No.

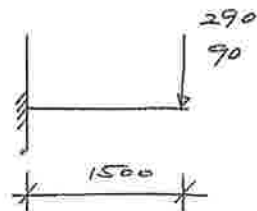
SA, PB.

Designed by

Sheet No.

Beam *4B2A/H1-G (300 x 1500 / 1700)*

Point load from *4B5/2-2a and 4B5/2a-8.*



PROKON

Software Consultants (Pty) Ltd

Internet: <http://www.prokon.com>

E-Mail: mail@prokon.com

Job Number

Job Title

Client

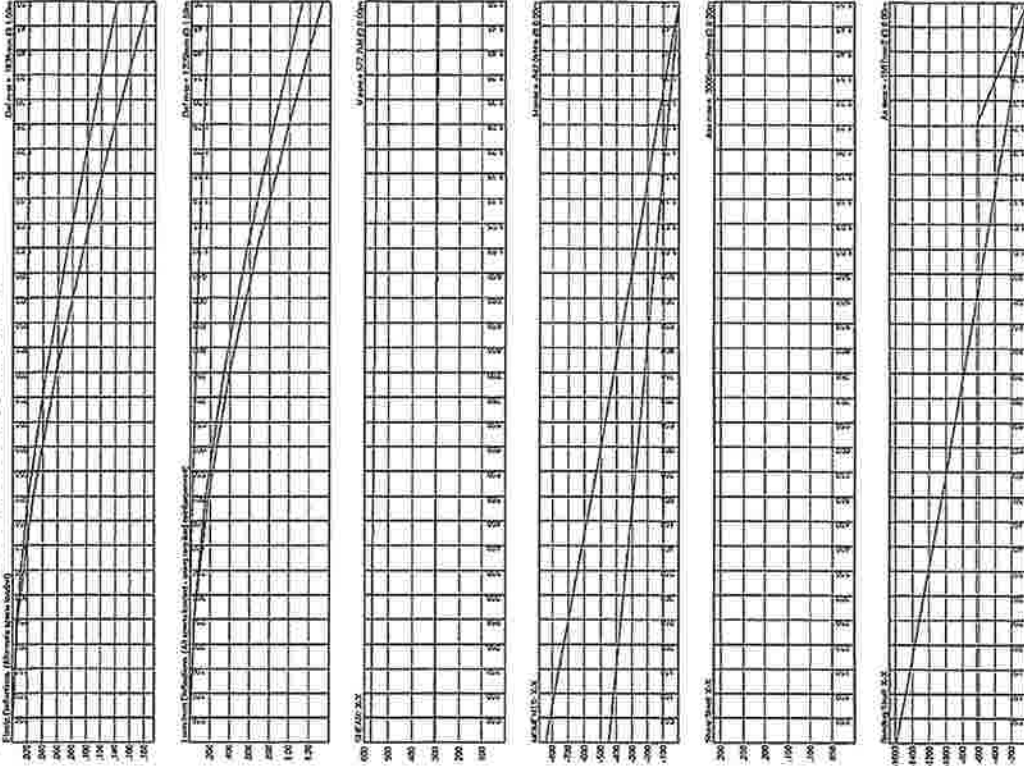
Calls by

Sheet

Checked by

Date

Deflection, Shear, Moment and Steel diagrams. Span 1



PROKON

Software Consultants (Pty) Ltd

Internet: <http://www.prokon.com>

E-Mail: mail@prokon.com

Job Number

Job Title

Client

Calls by

Sheet

Checked by

Date

Continuous Beam: Beam Design 4B2AA/H1-G (Code of Practice : BS8110 - 1997)

8/6/2007 11:08:00 AM

Input Tables

Fcu (MPa)	40
Fyk (MPa)	420
Fyk (MPa)	420
% Redistribution	0
Downward/Optimized redistr.	D
Cover to centre top steel (mm)	100
Cover to centre bot. steel (mm)	60
Dead Load Factor	1.4
Live Load Factor	1.6
Density of concrete (kN/m3)	24
% Live load permanent	25
Ø (Creep coefficient)	2
Ecs (Free shrinkage strain)	300E-6

Sec No.	Bw (mm)	D (mm)	Bf-top (mm)	Hf-top (mm)	Bf-bot (mm)	Hf-bot (mm)	Y-offset (mm)	Web offset	Flange offset
1	300	1500							

Span No	Section Length (m)	Sec No Left	Sec No Right
1	1.5	1	

Sup No.	Code C,F	Column Below D (mm)	B (mm)	H (mm)	Code F,P	Column above D (mm)	B (mm)	H (mm)
1	F							
2	C							

Case	Span	Wt (kNm)	Wright (kNm)	a (m)	b (m)	P (kN)	a (m)	M (kNm)	a (m)
D.L	1					280	1.5		
L	1					50	1.5		

Span	M left (kNm)	M right (kNm)

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C O N T I N U O U S B E A M V e r W2.2.23 - 27 Nov 2006

Title : Beam Design 4B2AA/H1-G
 Date File : 8/6/2007
 Date/Time : 8/6/2007
 Code of Practice : BS8110 - 1997

BENDING MOMENTS & REINFORCEMENT

SPAN 1

Position (m)	Min.Moment (kNm)	Max.Moment (kNm)	As-top (mm ²)	As-bot (mm ²)
0.000	-842.01	-447.15	1586.69	0.00
0.250	-699.31	-370.94	1317.79	0.00
0.500	-557.56	-295.40	1050.67	0.00
0.750	-416.75	-220.54	785.33	0.00
1.000	-276.89	-146.35	521.77	0.00
1.250	-137.97	-72.84	260.00	0.00
1.500	0.00	0.00	0.00	0.00

Shear Forces & REINFORCEMENT

SPAN 1

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm ² /mm)	Asv/Sv (nom) (mm ² /mm)
0.000	306.20	572.68	2160.00	0.00	0.30
0.250	303.50	568.90	2160.00	0.00	0.30
0.500	300.80	565.12	2160.00	0.00	0.30
0.750	298.10	561.34	2160.00	0.00	0.30
1.000	295.40	557.56	2160.00	0.00	0.30
1.250	292.70	553.78	2160.00	0.00	0.30
1.500	290.00	550.00	2100.00	0.00	0.30

COLUMN REACTIONS

Support	Dead Load (kN)	Live Load (kN)	Dead + Live (kN)	Ultimate (kN)
1	306.20	90.00	396.20	572.68
2	0.00	0.00	0.00	0.00

COLUMN MOMENTS (Ultimate limit state)

Columns below

Support No.	Bending Moment (kNm) Min.	Bending Moment (kNm) Max.	Shear Force (kN) Min.	Shear Force (kN) Max.
1	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00

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Columns above

Support No.	Bending Moment (kNm) Min.	Bending Moment (kNm) Max.	Shear Force (kN) Min.	Shear Force (kN) Max.
1	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00

ELASTIC DEFLECTIONS (Alternate spans loaded - positive downward)

Modulus of elasticity: $E_c = 20 + f_{cu}/5 = 28.0 \text{ Mpa}$

SPAN 1

Position (m)	Min.Deflection (mm)	Max.Deflection (mm)
0.000	-0.00	-0.00
0.250	0.01	0.01
0.500	0.02	0.03
0.750	0.04	0.06
1.000	0.07	0.10
1.250	0.11	0.14
1.500	0.14	0.18

LONGTERM DEFLECTIONS (All spans loaded - positive downward)

Symbols: DS : Shrinkage deflection.
 DL : Long-term deflection under permanent load.
 DI : Instantaneous deflection under short-term load.
 DT : Total long-term deflection.

SPAN 1

Position (m)	DS (mm)	DL (mm)	DI (mm)	DT (mm)
0.000	-0.00	0.00	0.00	-0.00
0.250	0.01	0.03	0.01	0.05
0.500	0.02	0.13	0.03	0.18
0.750	0.05	0.29	0.06	0.40
1.000	0.09	0.49	0.06	0.67
1.250	0.13	0.71	0.10	0.99
1.500	0.17	0.95	0.20	1.33

Total weight of beam = 16.20 kN



Site

Dubai Metro

Date

Job No

SA BB.



Designed by

Sheet No.

beam: 4B3 / 51 - H (200 x 1500)

loading:

dead load:

$$\text{slab: } (0.12 \times 24 + 1.12) \times 1.5 = 9 \text{ kN/m}$$

$$\text{sky light: } 1.15 \times 4.8 = 7.2 \text{ kN/m}$$

live load:

$$\text{slab: } 5 \times 1.5 = 7.5 \text{ kN/m}$$

$$\text{sky light: } 0.75 \times 4.8 = 3.6 \text{ kN/m}$$

4B3 / H - G (200 x 1500)

loading:

$$\text{dl: slab: } (0.12 \times 24 + 1.12) \times 0.4 = 2.4 \text{ kN/m}$$

$$\text{sky light: } 1.15 \times 5.5 = 8.25 \text{ kN/m}$$

$$\text{ll: } 5 \times 0.4 = 2 \text{ kN/m}$$

$$0.75 \times 5.5 = 4.125 \text{ kN/m}$$

point load from steel roof:

$$\text{dp: } 50 \text{ kN} \quad \text{up: } 27 \text{ kN}$$

4B3 / G - G1 (200 x 1500)

loading:

$$\text{dl: sky light: } 1.15 \times 6 = 9 \text{ kN/m}$$

$$\text{ll: } 0.75 \times 6 = 4.5 \text{ kN/m}$$

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site

Dubai Metro

Date

Job No.

JA. BB

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Sheet No.

point load from parapet wall.
DP: 21 kN EP: 45 kN

Continuous Beam: beam design for 4B3/J1-H to 4B3/G-G1 (Code of Practice BS8110 - 1997)
 8/4/2007 4:06:17 PM
 Date

Input Tables

Fcu (MPa)	40
Fyk (MPa)	460
Ftd (MPa)	460
% Redistribution	0
Downward/Optimized redistribution	D
Cover to centre top steel (mm)	40
Cover to centre bot steel (mm)	40
Dead Load Factor	1.4
Live Load Factor	1.6
Density of concrete (kN/m ³)	24
% Live load permanent	25
φ (Creep coefficient)	2
Ecs (Free shrinkage strain)	300E-6

Sec No.	BW (mm)	D (mm)	Bf-top (mm)	Hf-top (mm)	Bf-bot (mm)	Hf-bot (mm)	Y-offset (mm)	Web offset	Flange offset
1	200	1500							

Span No	Section Length (m)	Sec No Left	Sec No Right
1	4	1	
2	7.2	1	
3	1	1	

Sup No	Code C,F	Column Below D (mm)	B (mm)	H (m)	Code F,P	Column above D (mm)	B (mm)	H (m)	Code F,P
4	C								

Case D,L	Span	Wleft (kNm)	Wright (kNm)	a (m)	b (m)	P (kN)	a (m)	M (kNm)	a (m)
D	1	16.2							
L	1	11.1							
D	2	10.65				50	3.1		
L	2	6.125				27	3.1		
D	3	9				21	1		
L	3	4.5				4.5	1		

Span	M left (kNm)	M right (kNm)

Deflection, Shear, Moment and Steel diagrams. Spans 1-3



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CONTINUOUS BEAM Ver W2.2.23 - 27 Nov 2006

Title : beam design for 4B3/J1-H to 4B3/G-G1

Data File :

Date/Time : 8/4/2007

Code of Practice : BS8110 - 1997

BENDING MOMENTS & REINFORCEMENT

SPAN 1

Position (m)	Min.Moment (kNm)	Max.Moment (kNm)	As-Top (mm2)	As-bot (mm2)
0.000	0.00	0.00	0.00	0.00
0.250	-4.85	15.03	8.00	24.80
0.500	-11.16	26.90	18.41	44.39
0.750	-18.94	35.62	31.24	58.77
1.000	-28.17	41.18	46.48	67.94
1.250	-38.87	43.58	64.13	71.90
1.500	-51.03	42.82	84.20	70.65
1.750	-64.66	38.91	106.68	64.20
2.000	-79.75	31.84	131.57	52.53
2.250	-96.30	21.61	158.87	35.65
2.500	-114.31	8.22	188.59	13.57
2.750	-133.78	-8.32	220.72	0.00
3.000	-154.72	-28.02	255.26	0.00
3.250	-177.12	-50.88	292.22	0.00
3.500	-200.98	-76.90	331.59	0.00
3.750	-226.31	-106.07	373.37	0.00
4.000	-265.25	-138.40	437.62	0.00

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SPAN 2

Position (m)	Min.Moment (kNm)	Max.Moment (kNm)	As-top (mm2)	As-bot (mm2)
0.000	-265.25	-138.40	437.62	0.00
0.250	-211.49	-112.76	348.93	0.00
0.500	-159.91	-88.24	263.82	0.00
0.750	-110.50	-64.83	182.30	0.00
1.000	-63.26	-42.53	104.37	0.00
1.250	-21.35	-4.25	35.23	0.00
1.500	-1.29	38.99	2.13	64.33
1.750	17.66	80.06	0.00	132.09
2.000	35.49	118.96	0.00	196.26
2.250	52.21	155.68	0.00	256.85
2.500	67.81	190.23	0.00	313.84
2.750	82.29	222.60	0.00	367.25
3.000	95.66	252.80	0.00	417.08
3.250	100.70	264.27	0.00	436.00
3.500	100.00	263.12	0.00	434.11
3.600	98.20	259.80	0.00	428.63
3.850	95.27	254.31	0.00	419.57
4.100	91.23	246.64	0.00	406.92
4.350	86.08	236.80	0.00	390.68
4.600	79.81	224.78	0.00	370.85
4.850	72.42	210.59	0.00	347.44
5.100	63.92	194.22	0.00	320.43
5.350	54.30	175.68	0.00	289.84
5.600	43.57	154.97	0.00	255.67

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SPAN 3									
Position (m)	Min. Moment (kNm)	Max. Moment (kNm)	As-top (mm2)	As-bot (mm2)					
5.850	31.72	132.08	0.00	217.90					
6.100	18.76	107.01	0.00	176.55					
6.350	4.68	79.77	0.00	131.61					
6.600	-10.52	50.36	17.35	83.09					
6.850	-26.83	18.77	44.27	30.98					
7.100	-44.26	-14.99	73.02	0.00					
7.200	-51.54	-29.10	85.03	0.00					
SPAN 3									
Position (m)	Min. Moment (kNm)	Max. Moment (kNm)	As-top (mm2)	As-bot (mm2)					
0.000	-51.54	-29.10	85.03	0.00					
0.250	-35.85	-20.31	59.15	0.00					
0.500	-22.03	-12.52	36.35	0.00					
0.750	-10.08	-5.76	16.64	0.00					
1.000	0.00	0.00	0.00	0.00					
SHEAR FORCES & REINFORCEMENT									
SPAN 1									
Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)				
0.000	-16.47	66.44	1460.00	0.00	0.18				
0.250	-22.32	53.81	1339.52	0.00	0.18				
0.500	-28.17	41.18	669.76	0.00	0.18				
0.750	-34.02	28.55	446.51	0.00	0.18				
1.000	-39.87	15.92	334.88	0.00	0.18				
1.250	-45.72	3.29	267.90	0.00	0.18				
1.500	-51.57	-9.34	223.25	0.00	0.18				
1.750	-57.42	-21.97	191.36	0.00	0.18				
2.000	-66.31	-34.60	167.44	0.00	0.18				
2.250	-78.94	-47.23	191.36	0.00	0.18				
2.500	-91.57	-59.86	223.25	0.00	0.18				
2.750	-104.20	-72.49	267.90	0.00	0.18				
3.000	-116.83	-85.12	334.88	0.00	0.18				
3.250	-129.46	-92.52	446.51	0.00	0.18				
3.500	-142.09	-98.37	669.76	0.00	0.18				
3.750	-154.72	-104.22	1339.52	0.00	0.18				
4.000	-167.35	-110.07	1460.00	0.00	0.18				
SPAN 2									
Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)				
0.000	104.80	220.82	1460.00	0.00	0.18				
0.250	100.33	212.12	1339.52	0.00	0.18				
0.500	95.87	203.42	669.76	0.00	0.18				
0.750	91.41	194.72	446.51	0.00	0.18				
1.000	86.95	186.03	334.88	0.00	0.18				
1.250	82.48	177.33	267.90	0.00	0.18				
1.500	78.02	168.63	223.25	0.00	0.18				
1.750	73.56	159.93	191.36	0.00	0.18				
2.000	69.10	151.24	167.44	0.00	0.18				
2.250	64.63	142.54	148.84	0.00	0.18				
2.500	60.17	133.84	133.95	0.00	0.18				
2.750	55.71	125.14	121.77	0.01	0.18				

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SPAN 3

3.000	51.25	116.45	114.68	0.00	0.18
3.100	49.46	112.97	114.68	0.00	0.18
3.100	-1.66	-0.23	114.68	0.00	0.18
3.350	-10.36	-5.00	114.68	0.00	0.18
3.600	-19.06	-9.46	114.68	0.00	0.18
3.850	-27.75	-13.93	114.68	0.00	0.18
4.100	-36.45	-18.39	114.68	0.00	0.18
4.350	-45.15	-22.85	117.50	0.00	0.18
4.600	-53.85	-27.31	128.80	0.00	0.18
4.850	-62.54	-31.78	142.50	0.00	0.18
5.100	-71.24	-36.24	159.47	0.00	0.18
5.350	-79.94	-40.70	181.02	0.00	0.18
5.600	-88.64	-45.16	209.30	0.00	0.18
5.850	-97.33	-49.63	248.06	0.00	0.18
6.100	-106.03	-54.09	304.44	0.00	0.18
6.350	-114.73	-58.55	393.98	0.00	0.18
6.600	-123.43	-63.01	558.13	0.00	0.18
6.850	-132.12	-67.48	956.80	0.00	0.18
7.100	-140.82	-71.94	1460.00	0.00	0.18
7.200	-144.30	-73.72	1460.00	0.00	0.18

SPAN 3

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm2/mm)	Asv/Sv (nom) (mm2/mm)
0.000	37.20	66.48	1460.00	0.00	0.18
0.250	33.15	59.01	1339.52	0.00	0.18
0.500	29.10	51.54	1460.00	0.00	0.18
0.750	25.05	44.07	1460.00	0.00	0.18
1.000	21.00	36.60	1460.00	0.00	0.18

COLUMN REACTIONS

Support	Dead Load (kN)	Live Load (kN)	Dead + Live (kN)	Ultimate (kN)
1	15.24	8.37	23.61	66.44
2	184.59	80.20	264.79	386.74
3	109.50	35.93	145.42	210.78
4	0.00	0.00	0.00	0.00

COLUMN MOMENTS (Ultimate limit state)

Columns below

Support No.	Bending Moment (kNm)	Shear Force (kN)
	Min.	Max.
1	0.00	0.00
2	0.00	0.00
3	0.00	0.00
4	0.00	0.00

Columns above

Support No.	Bending Moment (kNm)	Shear Force (kN)
	Min.	Max.

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1

0.00

0.00

0.00

0.00

0.00

2

0.00

0.00

0.00

0.00

0.00

3

0.00

0.00

0.00

0.00

0.00

4

0.00

0.00

0.00

0.00

0.00

ELASTIC DEFLECTIONS (Alternate spans loaded - positive downward)

Modulus of elasticity: Ec = 20 + fcu/5 = 28.0 Mpa

SPAN 1

Position (m)

Min.Deflection (mm)

Max.Deflection (mm)

0.000

-0.00

-0.00

0.250

-0.01

0.00

0.500

-0.02

0.00

0.750

-0.03

-0.00

1.000

-0.04

-0.00

1.250

-0.04

-0.00

1.500

-0.05

-0.00

1.750

-0.06

-0.01

2.000

-0.06

-0.01

2.250

-0.07

-0.01

2.500

-0.07

-0.02

2.750

-0.07

-0.02

3.000

-0.06

-0.02

3.250

-0.05

-0.02

3.500

-0.04

-0.02

3.750

-0.02

-0.01

4.000

-0.00

-0.00

SPAN 2

Position (m)

Min.Deflection (mm)

Max.Deflection (mm)

0.000

-0.00

-0.00

0.250

0.02

0.03

0.500

0.03

0.07

0.750

0.06

0.10

1.000

0.08

0.15

1.250

0.11

0.19

1.500

0.14

0.24

1.750

0.16

0.28

2.000

0.19

0.32

2.250

0.21

0.36

2.500

0.24

0.40

2.750

0.26

0.43

3.000

0.27

0.45

3.100

0.28

0.46

3.100

0.28

0.46

3.350

0.29

0.47

3.600

0.29

0.48

3.600

0.29

0.48

3.850

0.29

0.48

4.100

0.29

0.47

4.350

0.28

0.46

4.600

0.27

0.44

4.850

0.25

0.41

5.100

0.23

0.38

ELASTIC DEFLECTIONS (Alternate spans loaded - positive downward)

Modulus of elasticity: $E_c = 20 + f_{cu}/5 = 28.0 \text{ Mpa}$

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5.350

0.21

0.35

5.600

0.18

0.31

5.850

0.16

0.27

6.100

0.13

0.22

6.350

0.10

0.17

6.600

0.07

0.12

6.850

0.04

0.07

7.100

0.01

0.02

7.200

0.00

0.00

SPAN 3

Position (m)

Min. Deflection (mm)

Max. Deflection (mm)

0.000

0.00

0.00

0.250

-0.05

-0.03

0.500

-0.10

-0.06

0.750

-0.15

-0.08

1.000

-0.20

-0.11

LONGTERM DEFLECTIONS (all spans loaded - positive downward)

Symbols: DS : Shrinkage deflection.

DL : Long-term deflection under permanent load.

DI : Instantaneous deflection under short-term load.

DT : Total long-term deflection.

SPAN 1

Position (m)

DS (mm)

DL (mm)

DI (mm)

DT (mm)

0.000

-0.00

0.00

0.00

-0.00

0.250

-0.00

-0.02

-0.00

-0.03

0.500

-0.01

-0.04

-0.01

-0.05

0.750

-0.01

-0.06

-0.01

-0.08

1.000

-0.02

-0.08

-0.01

-0.11

1.250

-0.02

-0.10

-0.02

-0.14

1.500

-0.02

-0.12

-0.02

-0.17

1.750

-0.03

-0.14

-0.02

-0.19

2.000

-0.03

-0.16

-0.03

-0.22

2.250

-0.03

-0.16

-0.03

-0.22

2.500

-0.04

-0.18

-0.03

-0.25

2.750

-0.04

-0.19

-0.03

-0.27

3.000

-0.04

-0.20

-0.04

-0.28

3.250

-0.04

-0.19

-0.04

-0.27

3.500

-0.03

-0.16

-0.04

-0.23

3.750

-0.02

-0.10

-0.02

-0.14

4.000

-0.00

0.00

0.00

-0.00

SPAN 2

Position (m)

DS (mm)

DL (mm)

DI (mm)

DT (mm)

0.000

-0.00

0.00

0.00

-0.00

0.250

0.03

0.15

0.04

0.22

0.500

0.07

0.33

0.10

0.49

0.750

0.11

0.53

0.16

0.79

1.000

0.15

0.74

0.22

1.11

PROKON				Job Number	Street
Software Consultants (Pty) Ltd				Job Title	
Internet: http://www.prokon.com				Client	
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1.250	0.20	0.96	0.28	1.44	
1.500	0.24	1.18	0.34	1.76	
1.750	0.28	1.40	0.40	2.09	
2.000	0.33	1.61	0.47	2.40	
2.250	0.37	1.81	0.53	2.71	
2.500	0.41	2.00	0.58	2.99	
2.750	0.44	2.16	0.63	3.23	
3.000	0.46	2.29	0.68	3.43	
3.100	0.47	2.33	0.69	3.49	
3.350	0.49	2.41	0.71	3.61	
3.600	0.50	2.45	0.73	3.67	
3.850	0.50	2.45	0.72	3.67	
4.100	0.49	2.41	0.71	3.62	
4.350	0.48	2.34	0.69	3.51	
4.600	0.46	2.24	0.65	3.35	
4.850	0.43	2.11	0.61	3.14	
5.100	0.39	1.94	0.55	2.89	
5.350	0.36	1.75	0.49	2.60	
5.600	0.31	1.54	0.43	2.29	
5.850	0.27	1.32	0.37	1.95	
6.100	0.22	1.08	0.30	1.60	
6.350	0.17	0.84	0.23	1.24	
6.600	0.12	0.59	0.16	0.88	
6.850	0.07	0.34	0.10	0.51	
7.100	0.02	0.10	0.03	0.15	
7.200	-0.00	-0.00	-0.00	-0.00	
SPAN 3					
Position (m)	DS (mm)	DL (mm)	DI (mm)	DT (mm)	
0.000	-0.00	0.00	0.00	-0.00	
0.250	-0.05	-0.24	-0.07	-0.36	
0.500	-0.10	-0.48	-0.14	-0.72	
0.750	-0.15	-0.72	-0.20	-1.07	
1.000	-0.20	-0.96	-0.27	-1.43	
Total weight of beam = 87.84 kN					

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site: Dubai Metro

Date:

Job No:

JA, BB.

Designed by

Sheet No

Beam ABG/2-2a (200 x 1850)
ABG/2a-3

loading:

DL: slab: $(0.2 \times 24 + 1.12) \times 1.2 = 7.2 \text{ kN/m}$

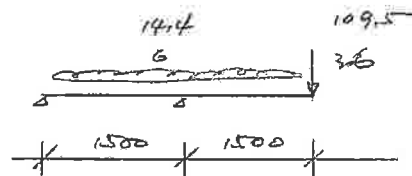
parapet wall: $0.2 \times 15 \times 24 = 7.2 \text{ kN/m}$

14.4 kN/m

LL: $5 \times 1.2 = 6 \text{ kN/m}$

Point load from ABG/H-G and ABG/G-G

DP: 109.5 kN CP: 36 kN



Continuous Beam: Beam Design 4BG/2-3 (Code of Practice : BS8110 - 1997)

8/4/2007 4:28:29 PM

Input Tables

Fcu (MPa)	40
Fy (MPa)	460
Fyv (MPa)	460
% Redistribution	0
Downward/Optimized redistr.	D
Cover to centre top steel(mm)	40
Cover to centre bot steel(mm)	40
Dead Load Factor	1.4
Live Load Factor	1.5
Density of concrete (kN/m ³)	24
% Live load permanent	25
Ø (Creep coefficient)	2
Ecs (Free shrinkage strain)	300E-6

Sec No.	Bw (mm)	D (mm)	BT-top (mm)	BT-bot (mm)	HT-top (mm)	HT-bot (mm)	Y-offset (mm)	Web offset	Flange offset
1	200	1800							

Span No	Section Length(m)	Sec No Left	Sec No Right
1	1.5	1	
2	1.5	1	

Sup No.	Code C,F	Column Below D(mm)	B(mm)	H(m)	Code F,P	Column above D(mm)	B(mm)	H(m)	Code F,P
3	C								

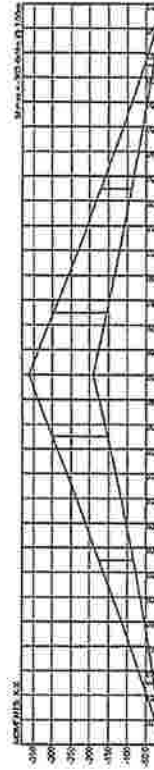
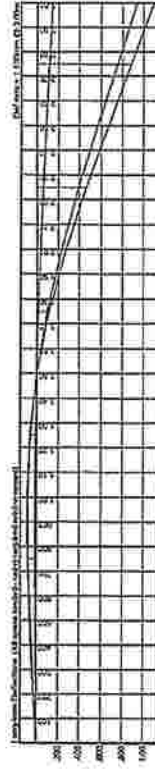
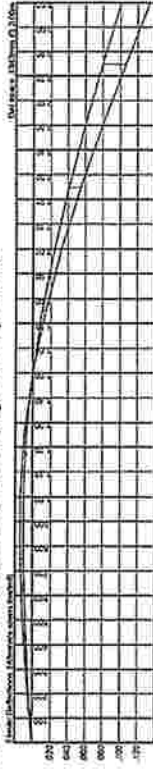
Case D.L	Span	Left (kNm)	Right (kNm)	a (m)	b (m)	P (kN)	a (m)	M (kNm)	a (m)
1	1	14.4							
2	2	14.4				108.5	1.5		
L	1	6							
	2	6				36	1.5		

Span	M left (kNm)	M right (kNm)

C01



Deflection, Shear, Moment and Steel diagrams. Spans 1,2



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CONTINUOUS BEAM Ver W2.2.23 - 27 Nov 2006			
Title : Beam Design 4BG/2-3			
Data File :			
Date/Time : 8/4/2007			
Code of Practice : BS8110 - 1997			

BENDING MOMENTS & REINFORCEMENT

SPAN 1

Position (m)	Min.Moment (kNm)	Max.Moment (kNm)	As-Top (mm ²)	As-Bot (mm ²)
0.000	0.00	0.00	0.00	0.00
0.250	-56.97	-25.16	77.97	0.00
0.500	-115.39	-52.93	157.92	0.00
0.750	-175.24	-83.31	239.84	0.00
1.000	-236.53	-116.32	323.72	0.00
1.250	-299.27	-151.94	409.58	0.00
1.500	-363.44	-190.17	497.41	0.00

SPAN 2

Position (m)	Min.Moment (kNm)	Max.Moment (kNm)	As-Top (mm ²)	As-Bot (mm ²)
0.000	-363.44	-190.17	497.41	0.00
0.250	-296.33	-154.88	405.56	0.00
0.500	-231.83	-121.02	317.28	0.00
0.750	-169.95	-88.61	232.59	0.00
1.000	-110.68	-57.63	151.48	0.00
1.250	-54.03	-28.10	73.95	0.00
1.500	0.00	0.00	0.00	0.00

SHEAR FORCES & REINFORCEMENT

SPAN 1

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm ² /mm)	Asv/Sv (nom) (mm ² /mm)
0.000	-225.01	-95.39	1760.00	0.00	0.18
0.250	-230.77	-105.85	1760.00	0.00	0.18
0.500	-236.53	-116.32	973.28	0.00	0.18
0.750	-242.29	-126.78	648.85	0.00	0.18
1.000	-252.76	-137.24	973.28	0.00	0.18
1.250	-263.22	-147.71	1760.00	0.00	0.18
1.500	-273.68	-158.17	1760.00	0.00	0.18

SPAN 2

Position (m)	Minimum Shear (kN)	Maximum Shear (kN)	Shear Capacity (kN)	Asv/Sv (mm ² /mm)	Asv/Sv (nom) (mm ² /mm)
0.000	144.06	273.68	1760.00	0.00	0.18
0.250	138.30	263.22	1760.00	0.00	0.18
0.500	132.54	252.76	973.28	0.00	0.18
0.750	126.78	242.29	1760.00	0.00	0.18

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1.000	121.02	231.83	1760.00	0.00	0.18
1.250	115.26	221.36	1760.00	0.00	0.18
1.500	109.50	210.90	1760.00	0.00	0.18

COLUMN REACTIONS

Support	Dead Load (kN)	Live Load (kN)	Dead + Live (kN)	Ultimate (kN)
1	-109.50	-36.00	-145.50	-225.01
2	288.12	90.00	378.12	547.37
3	0.00	0.00	0.00	0.00

COLUMN MOMENTS (Ultimate limit state)

Columns below

Support No.	Bending Moment (kNm)	Shear Force (kN)
	Min.	Max.
1	0.00	0.00
2	0.00	0.00
3	0.00	0.00

Columns above

Support No.	Bending Moment (kNm)	Shear Force (kN)
	Min.	Max.
1	0.00	0.00
2	0.00	0.00
3	0.00	0.00

ELASTIC DEFLECTIONS (Alternate spans loaded - positive downward)

Modulus of elasticity: $E_c = 20 + f_{cu}/5 = 28.0 \text{ Mpa}$

SPAN 1

Position (m)	Min.Deflection (mm)	Max.Deflection (mm)
0.000	-0.00	-0.00
0.250	-0.01	-0.00
0.500	-0.01	-0.01
0.750	-0.01	-0.01
0.750	-0.01	-0.01
1.000	-0.01	-0.01
1.250	-0.01	-0.01
1.500	-0.00	-0.00

SPAN 2

Position (m)	Min.Deflection (mm)	Max.Deflection (mm)
0.000	-0.00	-0.00
0.250	-0.01	0.01
0.500	0.02	0.03

PROKON				Job Number	Sheet
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0.750	0.04	0.05	0.05		
0.750	0.04	0.05	0.05		
1.000	0.06	0.08	0.08		
1.250	0.08	0.11	0.11		
1.500	0.10	0.13	0.13		
LONGTERM DEFLECTIONS (All spans loaded - positive downward)					
Symbols: DS : Shrinkage deflection.					
DL : Long-term deflection under permanent load.					
DI : Instantaneous deflection under short-term load.					
DT : Total long-term deflection.					
SPAN 1					
Position (m)	DS (mm)	DL (mm)	DI (mm)	DT (mm)	
0.000	-0.00	0.00	0.00	-0.00	
0.250	-0.00	-0.02	-0.00	-0.03	
0.500	-0.01	-0.05	-0.01	-0.06	
0.750	-0.01	-0.06	-0.01	-0.08	
1.000	-0.01	-0.06	-0.01	-0.08	
1.250	-0.01	-0.07	-0.01	-0.09	
1.500	-0.00	-0.05	-0.01	-0.07	
		-0.00	0.00	-0.00	
SPAN 2					
Position (m)	DS (mm)	DL (mm)	DI (mm)	DT (mm)	
0.000	-0.00	0.00	0.00	-0.00	
0.250	0.01	0.08	0.02	0.12	
0.500	0.04	0.20	0.04	0.28	
0.750	0.06	0.33	0.07	0.45	
1.000	0.09	0.33	0.07	0.46	
1.250	0.11	0.48	0.10	0.67	
1.500	0.14	0.64	0.13	0.88	
		0.80	0.16	1.10	
Total weight of beam = 25.92 kN					

4 Column Design

PROKON

Job Number
Job Title
Client
Software Consultants (Pty) Ltd
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Sheet

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Client
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Date

COL F5 L1

General column design by PROKON. (GenCol Ver W2.2.06 - 01 Dec 2006)
Design code : CP65 - 1999
Input tables
General design parameters:

Code	X/Radius or Bar dia. (mm)	Y (mm) Angle (°)
+	600.000	600.000
+	600.000	600.000
+	70.000	70.000
b	40	
+	530.000	70.000
b	40	
+	530.000	530.000
b	40	
+	70.000	530.000
b	40	
+	70.000	185.000
b	40.000	
+	630.000	185.000
b	40.000	
+	70.000	300.000
b	40.000	
+	530.000	300.000
b	40.000	
+	70.000	415.000
b	40.000	
+	530.000	415.000
b	40.000	
+	185.000	70.000
b	40.000	
+	185.000	530.000
b	40.000	
+	300.000	70.000
b	40.000	
+	300.000	530.000
b	40.000	
+	415.000	70.000
b	40.000	
+	415.000	530.000
b	40.000	

PROKON

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Design loads:

Load case	Designation	P (kN)	Mx top (kNm)	My top (kNm)	Mx bot (kNm)	My bot (kNm)
1		4230	536.4	0	0	0

Ultimate limit state design loads

General design parameters:

Given:
 $L_0 = 4.500 \text{ m}$
 $f_{cu} = 40 \text{ MPa}$
 $f_y = 460 \text{ MPa}$
 $A_c = 339894 \text{ mm}^2$

Assumptions:

- (1) The general conditions of clause 3.8.1 are applicable.
- (2) The specified design axial loads include the self-weight of the column.
- (3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:

- (1) The column design charts are constructed.
- (2) The design axis and design ultimate moment are determined.
- (3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:

At the top end: Condition 1 (fully fixed).

At the bottom end: Condition 1 (fully fixed).

This column is unbraced.

$\eta = 1.20$

Effective column height:

$l_e = l_0, L_0 = 5.400 \text{ m}$

Check if the column is slender:

$l_e/h = 9.1 < 10$

$\therefore$ The column is short.

Initial moments:

The initial end moments about the X-X axis:

$M_1 = \text{Smaller initial end moment} = 0.0 \text{ kNm}$

$M_2 = \text{Larger initial end moment} = 536.4 \text{ kNm}$

The initial moment near mid-height of the column :

$\therefore M_i = 0.4M_1 + 0.6M_2 \geq 0.4M_2 = 321.8 \text{ kNm}$

The initial end moments about the Y-Y axis:

$M_1 = \text{Smaller initial end moment} = 0.0 \text{ kNm}$

$M_2 = \text{Larger initial end moment} = 0.0 \text{ kNm}$

The initial moment near mid-height of the column :

$\therefore M_i = 0.4M_1 + 0.6M_2 \geq 0.4M_2 = 0.0 \text{ kNm}$

Table 3.21

3.8.1.3

3.8.3.2

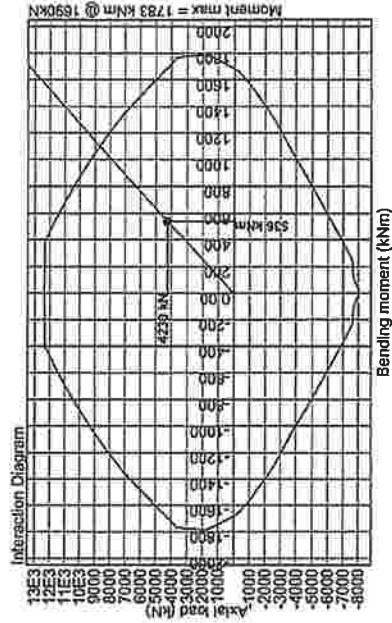
3.8.3.2

PROKON		Job Number	Street
Software Consultants (Pty) Ltd		Job Title	
Internet: http://www.prokon.com		Client	
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Design of column section for ULS:

The column is checked for applied moment about the design axis.
Through inspection: the critical section lies at the top end of the column.
The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
The safety factor for the critical load case 1 is 2.07

For bending about the design axis:

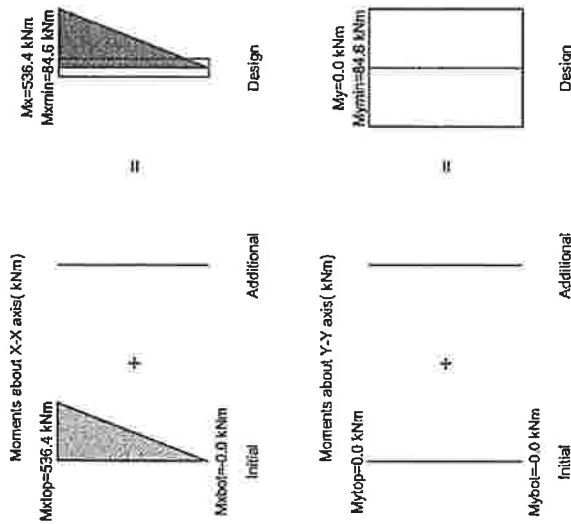


Moment distribution along the height of the column for bending about the design axis:
At the top, $M_x = 536.4 \text{ kNm}$
Near mid-height, $M_x = 321.8 \text{ kNm}$
At the bottom, $M_x = 84.6 \text{ kNm}$

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Design ultimate load and moment:

Design axial load:
 $P_u = 4230.0 \text{ kN}$

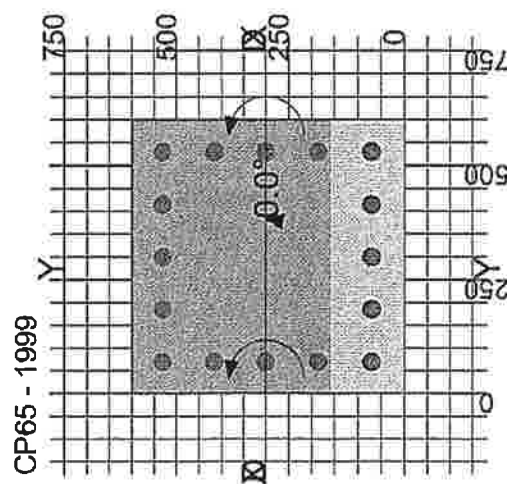


Check for minimum eccentricity:
Check that the eccentricity exceeds the minimum in the plane of bending:
Use $e_{min} = 20 \text{ mm}$
 $\therefore M_{min} = 84.6 \text{ kNm}$ about the X-X axis.

3.8.2.4

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Stresses at the top end of the column for the critical load case 1



Summary of design calculations:

Design results for all load cases:

Load case	Axis	N (kN)	M1 (kNm)	M2 (kNm)	M3 (kNm)	Madd (kNm)	Design	M (kNm)	M _r (kNm)	Safety factor
Load case 1	X-X	4230.0	0.0	536.4	321.8	0.0	Top	536.4	536.4	2.070
	Y-Y		0.0	0.0	0.0	0.0		0.0		

PROKON

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Job Number

068010.01

Job Title

Dubai Metro Proj.-JA building 8

Client

Drawn by

Ohm

Checked by

Dr Ravi

Date

7/23/2007

Sheet

1

COL F5 L1

General column design by PROKON. (GenCol Ver W2.2.06 - 01 Dec-2006)

Design code : CP65 - 1999

Input tables

General design parameters:

Code	X/Radius or Bar dia. (mm)	Y (mm)	Angle (°)
+	600.000		
	600.000	600.000	
	-600.000		
+	70.000	70.000	
b	40		
+	530.000	70.000	
b	40		
+	530.000	530.000	
b	40		
+	70.000	530.000	
b	40		
+	70.000	185.000	
b	40.000		
+	530.000	185.000	
b	40.000		
+	70.000	300.000	
b	40.000		
+	530.000	300.000	
b	40.000		
+	70.000	415.000	
b	40.000		
+	530.000	415.000	
b	40.000		
+	185.000	70.000	
b	40.000		
+	185.000	530.000	
b	40.000		
+	300.000	70.000	
b	40.000		
+	300.000	530.000	
b	40.000		
+	415.000	70.000	
b	40.000		
+	415.000	530.000	
b	40.000		

C13

PROKON

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Informatic: <http://www.prokon.com>

E-Mail: mail@prokon.com

Job Number

068010.01

Job Title

Dubai Metro Proj.-JA building 8

Client

Drawn by

Ohm

Checked by

Dr Ravi

Date

7/23/2007

Sheet

2

Design loads:

Load case	Designation	P (kN)	Mx top (kNm)	My top (kNm)	Mx bot (kNm)	My bot (kNm)
1		4230	536.4	0	0	0

Ultimate limit state design loads

General design parameters:

Given:

Lo = 4.500 m

f_{cu} = 40 MPa

f_y = 460 MPa

A_c = 339894 mm²

Assumptions:

(1) The general conditions of clause 3.8.1 are applicable.

(2) The specified design axial loads include the self-weight of the column.

(3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:

(1) The column design charts are constructed.

(2) The design axis and design ultimate moment are determined.

(3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:

At the top end: Condition 1 (fully fixed).

At the bottom end: Condition 1 (fully fixed).

The column is braced.

β = 0.75

Effective column height:

l_e = β · L_o = 3.375 m

Check if the column is slender:

l_e/h = 5.7 < 15

∴ The column is short.

Initial moments:

The initial end moments about the X-X axis:

M₁ = Smaller initial end moment = 0.0 kNm

M₂ = Larger initial end moment = 536.4 kNm

The initial moment near mid-height of the column :

∴ M_i = 0.4M₁ + 0.6M₂ ≥ 0.4M₂ = 321.8 kNm

The initial end moments about the Y-Y axis:

M₁ = Smaller initial end moment = 0.0 kNm

M₂ = Larger initial end moment = 0.0 kNm

The initial moment near mid-height of the column :

∴ M_i = 0.4M₁ + 0.6M₂ ≥ 0.4M₂ = 0.0 kNm

Table 3.21

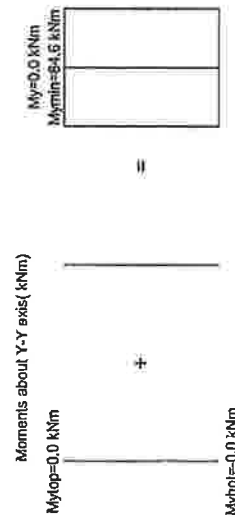
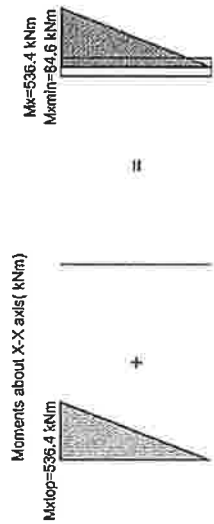
3.8.1.3

3.8.3.2

3.8.3.2

Design ultimate load and moment:

Design axial load:
 $P_u = 4230.0 \text{ kN}$



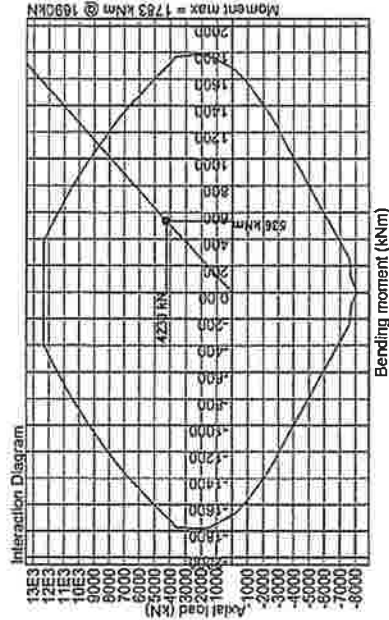
Check for minimum eccentricity:
 Check that the eccentricity exceeds the minimum in the plane of bending:
 Use $e_{min} = 20 \text{ mm}$
 $\therefore M_{min} = 84.6 \text{ kNm}$ about the X-X axis.

3.8.2.4

Design of column section for ULS:

The column is checked for applied moment about the design axis.
 Through inspection: the critical section lies at the top end of the column.
 The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
 The safety factor for the critical load case 1 is 2.07

For bending about the design axis:



Moment distribution along the height of the column for bending about the design axis:
 At the top, $M_x = 536.4 \text{ kNm}$
 Near mid-height, $M_x = 321.8 \text{ kNm}$
 At the bottom, $M_x = 84.6 \text{ kNm}$

PROKON

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COL D5 L1

General column design by PROKON. (GenCol Ver W2.2.06 - 01 Dec 2006)

Design code : CP65 - 1999

Input tables

General design parameters:

Code	X/Radius or Bar dia. (mm)	Y (mm) Angle (°)
a	600.000	500.000
b	650.000	70.000
c	70.000	70.000
d	40	70.000
e	530.000	70.000
f	40	530.000
g	40	530.000
h	70.000	530.000
i	40	185.000
j	40.000	185.000
k	530.000	300.000
l	40.000	300.000
m	40.000	415.000
n	530.000	415.000
o	185.000	70.000
p	40.000	530.000
q	40.000	70.000
r	300.000	70.000
s	40.000	530.000
t	40.000	70.000
u	415.000	70.000
v	40.000	530.000
w	415.000	530.000
x	40.000	

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Design loads:

Load case	Designation	P (kN)	Mx top (kNm)	My top (kNm)	Mx bot (kNm)	My bot (kNm)
1		5440	1335	0		

General design parameters:

Given:

Lo = 4.500 m

f_{cu} = 40 MPa

f_y = 460 MPa

A_c = 339894 mm²

Assumptions:

(1) The general conditions of clause 3.8.1 are applicable.

(2) The specified design axial loads include the self-weight of the column.

(3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:

(1) The column design charts are constructed.

(2) The design axis and design ultimate moment are determined.

(3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:

At the top end: Condition 1 (fully fixed).

At the bottom end: Condition 1 (fully fixed).

The column is unbraced

λ = 1.20

Effective column height:

l_e = λ · Lo = 5.400 m

Check if the column is slender:

l_e/h = 9.1 < 10

∴ The column is short.

Initial moments:

The initial end moments about the X-X axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 1335.0 kNm

The initial moment near mid-height of the column :

∴ Ml = 0.4M1 + 0.6M2 ≥ 0.4M2 = 801.0 kNm

The initial end moments about the Y-Y axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 0.0 kNm

The initial moment near mid-height of the column :

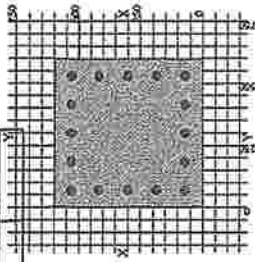
∴ Ml = 0.4M1 + 0.6M2 ≥ 0.4M2 = 0.0 kNm

Table 3.21

3.8.1.3

3.8.3.2

3.8.3.2



PROKON		Job Number	Sheet
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Design ultimate load and moment:
 $P_u = 5440.0 \text{ kN}$

Moments about X-X axis (kNm)

Initial: $M_{x\text{top}} = 1335.0 \text{ kNm}$, $M_{x\text{bot}} = 0.0 \text{ kNm}$

Additional: $M_x = 108.8 \text{ kNm}$

Design: $M_{x\text{top}} = 1335.0 \text{ kNm}$, $M_{x\text{bot}} = 108.8 \text{ kNm}$

Moments about Y-Y axis (kNm)

Initial: $M_{y\text{top}} = 0.0 \text{ kNm}$, $M_{y\text{bot}} = 0.0 \text{ kNm}$

Additional: $M_y = 108.8 \text{ kNm}$

Design: $M_{y\text{top}} = 108.8 \text{ kNm}$, $M_{y\text{bot}} = 108.8 \text{ kNm}$

Check for minimum eccentricity:
 Check that the eccentricity exceeds the minimum in the plane of bending:
 Use $e_{\min} = 20\text{mm}$
 $\therefore M_{\min} = 108.8 \text{ kNm}$ about the X-X axis.

3.8.2.4

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Design of column section for ULS:

The column is checked for applied moment about the design axis.
 Through inspection: the critical section lies at the top end of the column.
 The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
 The safety factor for the critical load case 1 is 1.12

For bending about the design axis:

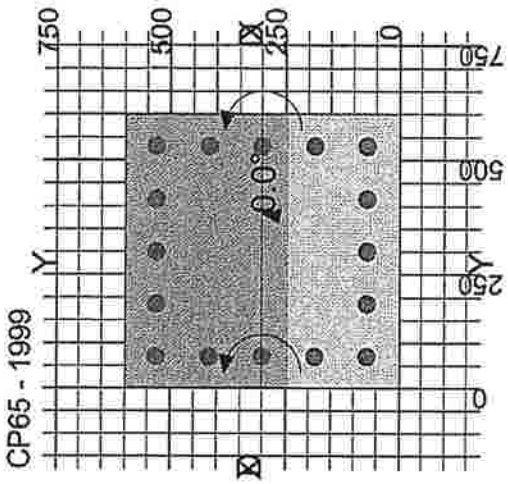
Interaction Diagram

Design point: $N = 5440 \text{ N}$, $M = 1335 \text{ kNm}$

Maximum moment: 1793 kNm @ 1890 kN

Moment distribution along the height of the column for bending about the design axis:
 At the top, $M_x = 1335.0 \text{ kNm}$
 Near mid-height, $M_x = 801.0 \text{ kNm}$
 At the bottom, $M_x = 108.8 \text{ kNm}$

Stresses at the top end of the column for the critical load case 1



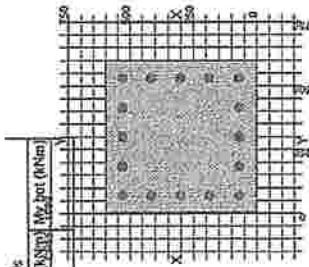
Summary of design calculations:

Design results for all load cases:

Load case	Axis	N (kN)	M1 (kNm)	M2 (kNm)	M1 (kNm)	M2 (kNm)	Madd (kNm)	Design	M (kNm)	M' (kNm)	Safety factor
Load case 1	X-X Y-Y	5440.0	0.0	1335.0	0.0	1335.0	0.0	Top	1335.0	1335.0	1.116

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COL D5 L1 General column design by PROKON. (GenCol Ver W2.2.06 - 01 Dec 2006) Design code : CP65 - 1999 Input tables General design parameters:																																																																																																																	
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Code</th> <th>Xi Radius or Bar dia. (mm)</th> <th>Y (mm) Angle (°)</th> </tr> </thead> <tbody> <tr><td>+</td><td>600.000</td><td></td></tr> <tr><td></td><td></td><td>600.000</td></tr> <tr><td>+</td><td>600.000</td><td></td></tr> <tr><td></td><td></td><td>70.000</td></tr> <tr><td>b</td><td>40</td><td></td></tr> <tr><td>+</td><td>530.000</td><td>70.000</td></tr> <tr><td>b</td><td>40</td><td></td></tr> <tr><td>+</td><td>530.000</td><td>530.000</td></tr> <tr><td>b</td><td>40</td><td></td></tr> <tr><td>+</td><td>70.000</td><td>530.000</td></tr> <tr><td>b</td><td>40</td><td></td></tr> <tr><td>+</td><td>70.000</td><td>185.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>530.000</td><td>185.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>70.000</td><td>300.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>530.000</td><td>300.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>70.000</td><td>415.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>530.000</td><td>415.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>185.000</td><td>70.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>185.000</td><td>530.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>300.000</td><td>70.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>300.000</td><td>530.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>415.000</td><td>70.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> <tr><td>+</td><td>415.000</td><td>530.000</td></tr> <tr><td>b</td><td>40.000</td><td></td></tr> </tbody> </table>						Code	Xi Radius or Bar dia. (mm)	Y (mm) Angle (°)	+	600.000				600.000	+	600.000				70.000	b	40		+	530.000	70.000	b	40		+	530.000	530.000	b	40		+	70.000	530.000	b	40		+	70.000	185.000	b	40.000		+	530.000	185.000	b	40.000		+	70.000	300.000	b	40.000		+	530.000	300.000	b	40.000		+	70.000	415.000	b	40.000		+	530.000	415.000	b	40.000		+	185.000	70.000	b	40.000		+	185.000	530.000	b	40.000		+	300.000	70.000	b	40.000		+	300.000	530.000	b	40.000		+	415.000	70.000	b	40.000		+	415.000	530.000	b	40.000	
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PROKON Software Consultants (Pty) Ltd Internet: http://www.prokon.com E-Mail: mail@prokon.com				Job Number: 068010.01 Job Title: Dubai Metro Proj-JA building 8 Client: Omm Checked by: Dr Ravi Date: 7/23/2007	Sheet: 2														
Design loads: <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Load case</th> <th>Designation</th> <th>P (kN)</th> <th>Mx top (kNm)</th> <th>Mx bot (kNm)</th> <th>My top (kNm)</th> <th>My bot (kNm)</th> </tr> </thead> <tbody> <tr> <td>1</td> <td></td> <td>5440</td> <td>1335</td> <td>0</td> <td></td> <td></td> </tr> </tbody> </table>						Load case	Designation	P (kN)	Mx top (kNm)	Mx bot (kNm)	My top (kNm)	My bot (kNm)	1		5440	1335	0		
Load case	Designation	P (kN)	Mx top (kNm)	Mx bot (kNm)	My top (kNm)	My bot (kNm)													
1		5440	1335	0															
General design parameters: Given: Lo = 4.500 m fcu = 40 MPa fy = 460 MPa Ac = 339894 mm ² Assumptions: (1) The general conditions of clause 3.8.1 are applicable. (2) The specified design axial loads include the self-weight of the column. (3) The design axial loads are taken constant over the height of the column.																			
Design approach: The column is designed using the following procedure: (1) The column design charts are constructed. (2) The design axis and design ultimate moment are determined. (3) The design axial force and moment capacity is checked on the relevant design chart.																			
Check column slenderness: End fixity and bracing for bending about the Design axis: At the top end: Condition 1 (fully fixed). At the bottom end: Condition 1 (fully fixed). The column is braced. $\beta = 0.75$ Effective column height: $l_e = \beta \cdot L_o = 3.375 \text{ m}$ Check if the column is slender: $l_e/h = 5.7 < 15$ $\therefore$ The column is short.																			
Initial moments: The initial end moments about the X-X axis: M1 = Smaller initial end moment = 0.0 kNm M2 = Larger initial end moment = 1335.0 kNm The initial moment near mid-height of the column : $\therefore Ml = 0.4M1 + 0.6M2 = 801.0 \text{ kNm}$ The initial end moments about the Y-Y axis: M1 = Smaller initial end moment = 0.0 kNm M2 = Larger initial end moment = 0.0 kNm The initial moment near mid-height of the column : $\therefore Ml = 0.4M1 + 0.6M2 = 0.0 \text{ kNm}$																			



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	Job Title	Dubai Metro Proj.-JA building 8		
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	Checked by	Ohm	Checked by	Dr Ravi
	Date	7/23/2007		

Design ultimate load and moment:

Design axial load:
 $P_u = 5440.0 \text{ kN}$

Moments about X-X axis (kNm)

Initial: $M_{\text{top}}=1335.0 \text{ kNm}$, $M_{\text{bot}}=0.0 \text{ kNm}$

Additional: $M_{\text{top}}=0.0 \text{ kNm}$, $M_{\text{bot}}=108.8 \text{ kNm}$

Design: $M_{\text{top}}=1335.0 \text{ kNm}$, $M_{\text{min}}=108.8 \text{ kNm}$

Moments about Y-Y axis (kNm)

Initial: $M_{\text{top}}=0.0 \text{ kNm}$, $M_{\text{bot}}=0.0 \text{ kNm}$

Additional: $M_{\text{top}}=0.0 \text{ kNm}$, $M_{\text{bot}}=108.8 \text{ kNm}$

Design: $M_{\text{top}}=0.0 \text{ kNm}$, $M_{\text{min}}=108.8 \text{ kNm}$

Check for minimum eccentricity:
 Check that the eccentricity exceeds the minimum in the plane of bending:
 Use $e_{\text{min}} = 20\text{mm}$
 $\therefore M_{\text{min}} = 108.8 \text{ kNm}$ about the X-X axis.

3.3.2.4

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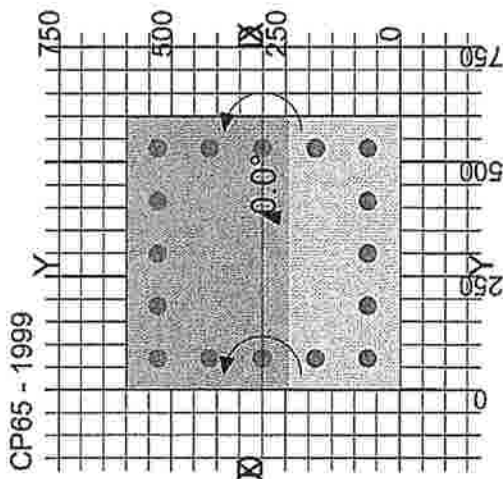
Design of column section for ULS:

The column is checked for applied moment about the design axis.
 Through inspection: the critical section lies at the top end of the column.
 The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
 The safety factor for the critical load case 1 is 1.12

For bending about the design axis:

Moment distribution along the height of the column for bending about the design axis:
 At the top, $M_x = 1335.0 \text{ kNm}$
 Near mid-height, $M_x = 801.0 \text{ kNm}$
 At the bottom, $M_x = 108.8 \text{ kNm}$

Stresses at the top end of the column for the critical load case 1



Summary of design calculations:

Design results for all load cases:

Load case	Axis	N (kN)	M1 (kNm)	M2 (kNm)	M3 (kNm)	M4 (kNm)	M5 (kNm)	M6 (kNm)	M7 (kNm)	M8 (kNm)	Safety factor
Load case 1	X-X Y-Y	5440.0 0.0	0.0 0.0	1335.0 0.0	601.0 0.0	0.0 0.0	0.0 0.0	1335.0 0.0	0.0 0.0	1335.0 0.0	1.116

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COL C5 L1

General column design by PROKON. (GenCol Ver W2.2.06 - 01 Dec 2006)

Design code : CP65 - 1999

Input tables

General design parameters:

Code	X/Radius or Bar dia. (mm)	Y (mm)	Angle (°)
a	600.000	800.000	
b	400.000	70.000	
b	40	70.000	
a	530.000	70.000	
b	40	530.000	
a	630.000	530.000	
b	40	70.000	
a	70.000	530.000	
b	40	70.000	
a	70.000	185.000	
b	40.000	185.000	
a	530.000	300.000	
b	40.000	300.000	
a	70.000	415.000	
b	40.000	415.000	
a	185.000	70.000	
b	40.000	70.000	
a	185.000	530.000	
b	40.000	530.000	
a	415.000	70.000	
b	40.000	70.000	
a	415.000	530.000	
b	40.000	530.000	

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Design loads:

Load case	Designation	P (kN)	Mx top (kNm)	My top (kNm)	Mx bot (kNm)	My bot (kNm)
1		5440	1133	0		

General design parameters:

Given:

Lo = 4.500 m

f_{cu} = 40 MPa

f_y = 460 MPa

A_c = 339894 mm²

Assumptions:

(1) The general conditions of clause 3.8.1 are applicable.

(2) The specified design axial loads include the self-weight of the column.

(3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:

(1) The column design charts are constructed.

(2) The design axis and design ultimate moment are determined.

(3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:

At the top end: Condition 1 (fully fixed).

At the bottom end: Condition 1 (fully fixed).

The column is unbraced.

l_b = 1.20

Effective column height:

l_e = l_b · Lo = 5.400 m

Check if the column is slender:

l_e/h = 9.1 < 10

∴ The column is short.

Initial moments:

The initial end moments about the X-X axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 1133.0 kNm

The initial moment near mid-height of the column :

∴ M_l = 0.4M1 + 0.6M2 = 679.8 kNm

The initial end moments about the Y-Y axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 0.0 kNm

The initial moment near mid-height of the column :

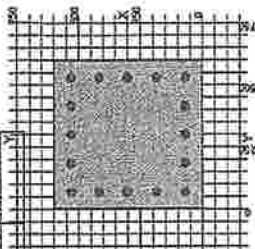
∴ M_l = 0.4M1 + 0.6M2 = 0.0 kNm

Table 3.21

3.B.1.3

3.8.3.2

3.8.3.2



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Design ultimate load and moment:

Design axial load:

$P_d = 5440.0 \text{ kN}$

Mx(top)=1133.0 kNm

Mx(bot)=0.0 kNm

Mx(top)=1133.0 kNm

Mx(min)=108.8 kNm

Initial

Additional

Design

My(top)=0.0 kNm

My(bot)=0.0 kNm

My(top)=0.0 kNm

My(min)=108.8 kNm

Initial

Additional

Design

Check for minimum eccentricity:

Check that the eccentricity exceeds the minimum in the plane of bending:

Use $e_{min} = 20\text{mm}$

$\therefore M_{min} = 108.8 \text{ kNm}$ about the X-X axis.

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Internet: <http://www.prokon.com>

E-Mail: mail@prokon.com

Design of column section for ULS:

The column is checked for applied moment about the design axis.

Through inspection, the critical section lies at the top end of the column.

The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis

The safety factor for the critical load case 1 is 1.24

For bending about the design axis:

Interaction Diagram

13E3

12E3

11E3

10E3

9000

8000

7000

6000

5000

4000

3000

2000

1000

0

-1000

-2000

-3000

-4000

-5000

-6000

-7000

-8000

13000

12000

11000

10000

9000

8000

7000

6000

5000

4000

3000

2000

1000

0

-1000

-2000

-3000

-4000

-5000

-6000

-7000

-8000

Axial load (kN)

Bending moment (kNm)

Moment max = 1783 kNm @ 1690kN

5440 kN

Moment distribution along the height of the column for bending about the design axis:

At the top, $M_x = 1133.0 \text{ kNm}$

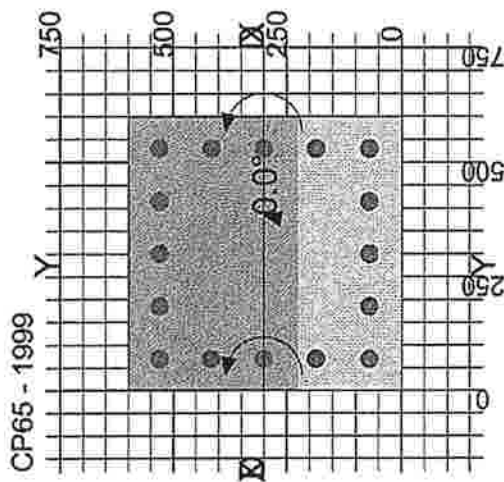
Near mid-height, $M_x = 679.8 \text{ kNm}$

At the bottom, $M_x = 108.8 \text{ kNm}$

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Stresses at the top end of the column for the critical load case 1



Summary of design calculations:

Design results for all load cases:

Load case	Axial	N (kN)	M1 (kNm)	M2 (kNm)	M3 (kNm)	M4 (kNm)	Design	M (kNm)	M' (kNm)	Safety factor
Load case 1	X-X	5440.0	0.0	1133.0	679.8	0.0	Top	1133.0	1133.0	1.241
	Y-Y		0.0	0.0	0.0	0.0				

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		Date	7/23/2007

COL C5 L1

General column design by PROKON, (GenCol Ver.W2.2.06 - 01 Dec 2006)

Design code : CP65 - 1999

Input tables

General design parameters:

Code	X/Radius or Bar dia. (mm)	Y (mm)	Angle (°)
a	600.000		
b	40.000	600.000	
c	40.000	70.000	
d	40.000	70.000	
e	40.000	70.000	
f	40.000	70.000	
g	40.000	70.000	
h	40.000	70.000	
i	40.000	70.000	
j	40.000	70.000	
k	40.000	70.000	
l	40.000	70.000	
m	40.000	70.000	
n	40.000	70.000	
o	40.000	70.000	
p	40.000	70.000	
q	40.000	70.000	
r	40.000	70.000	
s	40.000	70.000	
t	40.000	70.000	
u	40.000	70.000	
v	40.000	70.000	
w	40.000	70.000	
x	40.000	70.000	
y	40.000	70.000	
z	40.000	70.000	

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Design loads:

Load case	Designation	P (kN)	Max top (kNm)	My top (kNm)	Max bot (kNm)	My bot (kNm)
1		5440	1133	0	0	0

Ultimate limit state design loads

General design parameters:

Given:

Lo = 4.500 m

f_{cu} = 40 MPa

f_y = 460 MPa

Ac = 339894 mm²

Assumptions:

(1) The general conditions of clause 3.8.1 are applicable.

(2) The specified design axial loads include the self-weight of the column.

(3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:

(1) The column design charts are constructed.

(2) The design axis and design ultimate moment are determined.

(3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:

At the top end: Condition 1 (fully fixed).

At the bottom end: Condition 1 (fully fixed).

The column is braced.

$\beta = 0.75$

Effective column height:

le = $\beta \cdot Lo = 3.375$ m

Check if the column is slender:

le/h = 5.7 < 15

$\therefore$ The column is short.

Initial moments:

The initial end moments about the X-X axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 1133.0 kNm

The initial moment near mid-height of the column :

$\therefore Ml = 0.4M1 + 0.6M2 \geq 0.4M2 = 679.8$ kNm

The initial end moments about the Y-Y axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 0.0 kNm

The initial moment near mid-height of the column :

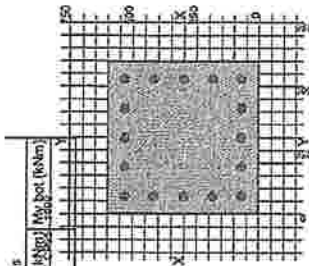
$\therefore Ml = 0.4M1 + 0.6M2 \geq 0.4M2 = 0.0$ kNm

Table 3.21

3.8.1.3

3.8.3.2

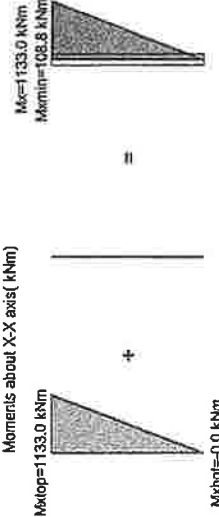
3.8.3.2



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Design ultimate load and moment:
 Design axial load:
 $P_u = 5440.0 \text{ kN}$

Moments about X-X axis (kNm)

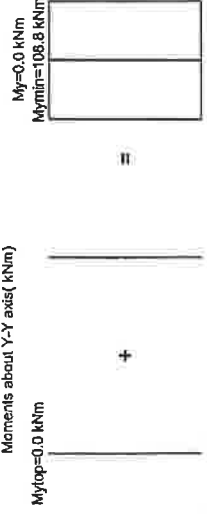


Initial

Additional

Design

Moments about Y-Y axis (kNm)



Initial

Additional

Design

Check for minimum eccentricity:
 Check that the eccentricity exceeds the minimum in the plane of bending:
 Use $e_{min} = 20\text{mm}$
 $\therefore M_{min} = 108.8 \text{ kNm}$ about the X-X axis.

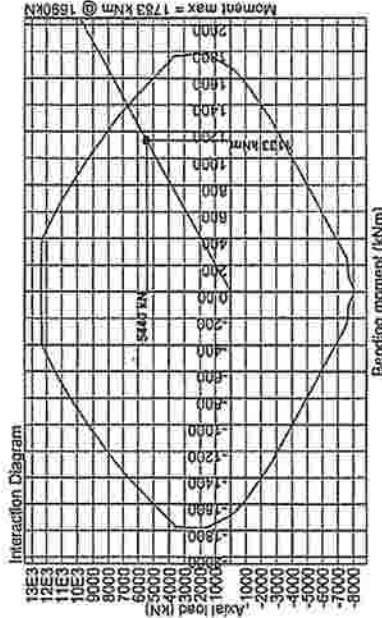
3.6.2.4

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Design of column section for ULS:

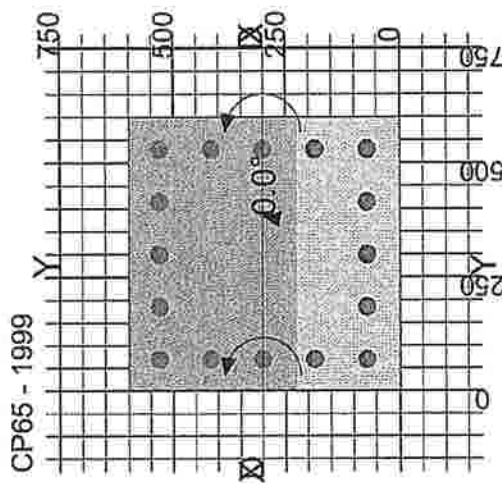
The column is checked for applied moment about the design axis.
 Through inspection: the critical section lies at the top end of the column.
 The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
 The safety factor for the critical load case 1 is 1.24

For bending about the design axis:



Moment distribution along the height of the column for bending about the design axis:
 At the top, $M_x = 1133.0 \text{ kNm}$
 Near mid-height, $M_x = 679.8 \text{ kNm}$
 At the bottom, $M_x = 108.8 \text{ kNm}$

Stresses at the top end of the column for the critical load case 1



Summary of design calculations:

Design results for all load cases:

Load case	Axis	N (kN)	M1 (kNm)	M2 (kNm)	M3 (kNm)	M4 (kNm)	M5 (kNm)	M6 (kNm)	M7 (kNm)	M8 (kNm)	M9 (kNm)	M10 (kNm)	M11 (kNm)	M12 (kNm)	M13 (kNm)	M14 (kNm)	M15 (kNm)	M16 (kNm)	M17 (kNm)	M18 (kNm)	M19 (kNm)	M20 (kNm)	M21 (kNm)	M22 (kNm)	M23 (kNm)	M24 (kNm)	M25 (kNm)	M26 (kNm)	M27 (kNm)	M28 (kNm)	M29 (kNm)	M30 (kNm)	M31 (kNm)	M32 (kNm)	M33 (kNm)	M34 (kNm)	M35 (kNm)	M36 (kNm)	M37 (kNm)	M38 (kNm)	M39 (kNm)	M40 (kNm)	M41 (kNm)	M42 (kNm)	M43 (kNm)	M44 (kNm)	M45 (kNm)	M46 (kNm)	M47 (kNm)	M48 (kNm)	M49 (kNm)	M50 (kNm)	M51 (kNm)	M52 (kNm)	M53 (kNm)	M54 (kNm)	M55 (kNm)	M56 (kNm)	M57 (kNm)	M58 (kNm)	M59 (kNm)	M60 (kNm)	M61 (kNm)	M62 (kNm)	M63 (kNm)	M64 (kNm)	M65 (kNm)	M66 (kNm)	M67 (kNm)	M68 (kNm)	M69 (kNm)	M70 (kNm)	M71 (kNm)	M72 (kNm)	M73 (kNm)	M74 (kNm)	M75 (kNm)	M76 (kNm)	M77 (kNm)	M78 (kNm)	M79 (kNm)	M80 (kNm)	M81 (kNm)	M82 (kNm)	M83 (kNm)	M84 (kNm)	M85 (kNm)	M86 (kNm)	M87 (kNm)	M88 (kNm)	M89 (kNm)	M90 (kNm)	M91 (kNm)	M92 (kNm)	M93 (kNm)	M94 (kNm)	M95 (kNm)	M96 (kNm)	M97 (kNm)	M98 (kNm)	M99 (kNm)	M100 (kNm)	M101 (kNm)	M102 (kNm)	M103 (kNm)	M104 (kNm)	M105 (kNm)	M106 (kNm)	M107 (kNm)	M108 (kNm)	M109 (kNm)	M110 (kNm)	M111 (kNm)	M112 (kNm)	M113 (kNm)	M114 (kNm)	M115 (kNm)	M116 (kNm)	M117 (kNm)	M118 (kNm)	M119 (kNm)	M120 (kNm)	M121 (kNm)	M122 (kNm)	M123 (kNm)	M124 (kNm)	M125 (kNm)	M126 (kNm)	M127 (kNm)	M128 (kNm)	M129 (kNm)	M130 (kNm)	M131 (kNm)	M132 (kNm)	M133 (kNm)	M134 (kNm)	M135 (kNm)	M136 (kNm)	M137 (kNm)	M138 (kNm)	M139 (kNm)	M140 (kNm)	M141 (kNm)	M142 (kNm)	M143 (kNm)	M144 (kNm)	M145 (kNm)	M146 (kNm)	M147 (kNm)	M148 (kNm)	M149 (kNm)	M150 (kNm)	M151 (kNm)	M152 (kNm)	M153 (kNm)	M154 (kNm)	M155 (kNm)	M156 (kNm)	M157 (kNm)	M158 (kNm)	M159 (kNm)	M160 (kNm)	M161 (kNm)	M162 (kNm)	M163 (kNm)	M164 (kNm)	M165 (kNm)	M166 (kNm)	M167 (kNm)	M168 (kNm)	M169 (kNm)	M170 (kNm)	M171 (kNm)	M172 (kNm)	M173 (kNm)	M174 (kNm)	M175 (kNm)	M176 (kNm)	M177 (kNm)	M178 (kNm)	M179 (kNm)	M180 (kNm)	M181 (kNm)	M182 (kNm)	M183 (kNm)	M184 (kNm)	M185 (kNm)	M186 (kNm)	M187 (kNm)	M188 (kNm)	M189 (kNm)	M190 (kNm)	M191 (kNm)	M192 (kNm)	M193 (kNm)	M194 (kNm)	M195 (kNm)	M196 (kNm)	M197 (kNm)	M198 (kNm)	M199 (kNm)	M200 (kNm)	M201 (kNm)	M202 (kNm)	M203 (kNm)	M204 (kNm)	M205 (kNm)	M206 (kNm)	M207 (kNm)	M208 (kNm)	M209 (kNm)	M210 (kNm)	M211 (kNm)	M212 (kNm)	M213 (kNm)	M214 (kNm)	M215 (kNm)	M216 (kNm)	M217 (kNm)	M218 (kNm)	M219 (kNm)	M220 (kNm)	M221 (kNm)	M222 (kNm)	M223 (kNm)	M224 (kNm)	M225 (kNm)	M226 (kNm)	M227 (kNm)	M228 (kNm)	M229 (kNm)	M230 (kNm)	M231 (kNm)	M232 (kNm)	M233 (kNm)	M234 (kNm)	M235 (kNm)	M236 (kNm)	M237 (kNm)	M238 (kNm)	M239 (kNm)	M240 (kNm)	M241 (kNm)	M242 (kNm)	M243 (kNm)	M244 (kNm)	M245 (kNm)	M246 (kNm)	M247 (kNm)	M248 (kNm)	M249 (kNm)	M250 (kNm)	M251 (kNm)	M252 (kNm)	M253 (kNm)	M254 (kNm)	M255 (kNm)	M256 (kNm)	M257 (kNm)	M258 (kNm)	M259 (kNm)	M260 (kNm)	M261 (kNm)	M262 (kNm)	M263 (kNm)	M264 (kNm)	M265 (kNm)	M266 (kNm)	M267 (kNm)	M268 (kNm)	M269 (kNm)	M270 (kNm)	M271 (kNm)	M272 (kNm)	M273 (kNm)	M274 (kNm)	M275 (kNm)	M276 (kNm)	M277 (kNm)	M278 (kNm)	M279 (kNm)	M280 (kNm)	M281 (kNm)	M282 (kNm)	M283 (kNm)	M284 (kNm)	M285 (kNm)	M286 (kNm)	M287 (kNm)	M288 (kNm)	M289 (kNm)	M290 (kNm)	M291 (kNm)	M292 (kNm)	M293 (kNm)	M294 (kNm)	M295 (kNm)	M296 (kNm)	M297 (kNm)	M298 (kNm)	M299 (kNm)	M300 (kNm)	M301 (kNm)	M302 (kNm)	M303 (kNm)	M304 (kNm)	M305 (kNm)	M306 (kNm)	M307 (kNm)	M308 (kNm)	M309 (kNm)	M310 (kNm)	M311 (kNm)	M312 (kNm)	M313 (kNm)	M314 (kNm)	M315 (kNm)	M316 (kNm)	M317 (kNm)	M318 (kNm)	M319 (kNm)	M320 (kNm)	M321 (kNm)	M322 (kNm)	M323 (kNm)	M324 (kNm)	M325 (kNm)	M326 (kNm)	M327 (kNm)	M328 (kNm)	M329 (kNm)	M330 (kNm)	M331 (kNm)	M332 (kNm)	M333 (kNm)	M334 (kNm)	M335 (kNm)	M336 (kNm)	M337 (kNm)	M338 (kNm)	M339 (kNm)	M340 (kNm)	M341 (kNm)	M342 (kNm)	M343 (kNm)	M344 (kNm)	M345 (kNm)	M346 (kNm)	M347 (kNm)	M348 (kNm)	M349 (kNm)	M350 (kNm)	M351 (kNm)	M352 (kNm)	M353 (kNm)	M354 (kNm)	M355 (kNm)	M356 (kNm)	M357 (kNm)	M358 (kNm)	M359 (kNm)	M360 (kNm)	M361 (kNm)	M362 (kNm)	M363 (kNm)	M364 (kNm)	M365 (kNm)	M366 (kNm)	M367 (kNm)	M368 (kNm)	M369 (kNm)	M370 (kNm)	M371 (kNm)	M372 (kNm)	M373 (kNm)	M374 (kNm)	M375 (kNm)	M376 (kNm)	M377 (kNm)	M378 (kNm)	M379 (kNm)	M380 (kNm)	M381 (kNm)	M382 (kNm)	M383 (kNm)	M384 (kNm)	M385 (kNm)	M386 (kNm)	M387 (kNm)	M388 (kNm)	M389 (kNm)	M390 (kNm)	M391 (kNm)	M392 (kNm)	M393 (kNm)	M394 (kNm)	M395 (kNm)	M396 (kNm)	M397 (kNm)	M398 (kNm)	M399 (kNm)	M400 (kNm)	M401 (kNm)	M402 (kNm)	M403 (kNm)	M404 (kNm)	M405 (kNm)	M406 (kNm)	M407 (kNm)	M408 (kNm)	M409 (kNm)	M410 (kNm)	M411 (kNm)	M412 (kNm)	M413 (kNm)	M414 (kNm)	M415 (kNm)	M416 (kNm)	M417 (kNm)	M418 (kNm)	M419 (kNm)	M420 (kNm)	M421 (kNm)	M422 (kNm)	M423 (kNm)	M424 (kNm)	M425 (kNm)	M426 (kNm)	M427 (kNm)	M428 (kNm)	M429 (kNm)	M430 (kNm)	M431 (kNm)	M432 (kNm)	M433 (kNm)	M434 (kNm)	M435 (kNm)	M436 (kNm)	M437 (kNm)	M438 (kNm)	M439 (kNm)	M440 (kNm)	M441 (kNm)	M442 (kNm)	M443 (kNm)	M444 (kNm)	M445 (kNm)	M446 (kNm)	M447 (kNm)	M448 (kNm)	M449 (kNm)	M450 (kNm)	M451 (kNm)	M452 (kNm)	M453 (kNm)	M454 (kNm)	M455 (kNm)	M456 (kNm)	M457 (kNm)	M458 (kNm)	M459 (kNm)	M460 (kNm)	M461 (kNm)	M462 (kNm)	M463 (kNm)	M464 (kNm)	M465 (kNm)	M466 (kNm)	M467 (kNm)	M468 (kNm)	M469 (kNm)	M470 (kNm)	M471 (kNm)	M472 (kNm)	M473 (kNm)	M474 (kNm)	M475 (kNm)	M476 (kNm)	M477 (kNm)	M478 (kNm)	M479 (kNm)	M480 (kNm)	M481 (kNm)	M482 (kNm)	M483 (kNm)	M484 (kNm)	M485 (kNm)	M486 (kNm)	M487 (kNm)	M488 (kNm)	M489 (kNm)	M490 (kNm)	M491 (kNm)	M492 (kNm)	M493 (kNm)	M494 (kNm)	M495 (kNm)	M496 (kNm)	M497 (kNm)	M498 (kNm)	M499 (kNm)	M500 (kNm)	M501 (kNm)	M502 (kNm)	M503 (kNm)	M504 (kNm)	M505 (kNm)	M506 (kNm)	M507 (kNm)	M508 (kNm)	M509 (kNm)	M510 (kNm)	M511 (kNm)	M512 (kNm)	M513 (kNm)	M514 (kNm)	M515 (kNm)	M516 (kNm)	M517 (kNm)	M518 (kNm)	M519 (kNm)	M520 (kNm)	M521 (kNm)	M522 (kNm)	M523 (kNm)	M524 (kNm)	M525 (kNm)	M526 (kNm)	M527 (kNm)	M528 (kNm)	M529 (kNm)	M530 (kNm)	M531 (kNm)	M532 (kNm)	M533 (kNm)	M534 (kNm)	M535 (kNm)	M536 (kNm)	M537 (kNm)	M538 (kNm)	M539 (kNm)	M540 (kNm)	M541 (kNm)	M542 (kNm)	M543 (kNm)	M544 (kNm)	M545 (kNm)	M546 (kNm)	M547 (kNm)	M548 (kNm)	M549 (kNm)	M550 (kNm)	M551 (kNm)	M552 (kNm)	M553 (kNm)	M554 (kNm)	M555 (kNm)	M556 (kNm)	M557 (kNm)	M558 (kNm)	M559 (kNm)	M560 (kNm)	M561 (kNm)	M562 (kNm)	M563 (kNm)	M564 (kNm)	M565 (kNm)	M566 (kNm)	M567 (kNm)	M568 (kNm)	M569 (kNm)	M570 (kNm)	M571 (kNm)	M572 (kNm)	M573 (kNm)	M574 (kNm)	M575 (kNm)	M576 (kNm)	M577 (kNm)	M578 (kNm)	M579 (kNm)	M580 (kNm)	M581 (kNm)	M582 (kNm)	M583 (kNm)	M584 (kNm)	M585 (kNm)	M586 (kNm)	M587 (kNm)	M588 (kNm)	M589 (kNm)	M590 (kNm)	M591 (kNm)	M592 (kNm)	M593 (kNm)	M594 (kNm)	M595 (kNm)	M596 (kNm)	M597 (kNm)	M598 (kNm)	M599 (kNm)	M600 (kNm)	M601 (kNm)	M602 (kNm)	M603 (kNm)	M604 (kNm)	M605 (kNm)	M606 (kNm)	M607 (kNm)	M608 (kNm)	M609 (kNm)	M610 (kNm)	M611 (kNm)	M612 (kNm)	M613 (kNm)	M614 (kNm)	M615 (kNm)	M616 (kNm)	M617 (kNm)	M618 (kNm)	M619 (kNm)	M620 (kNm)	M621 (kNm)	M622 (kNm)	M623 (kNm)	M624 (kNm)	M625 (kNm)	M626 (kNm)	M627 (kNm)	M628 (kNm)	M629 (kNm)	M630 (kNm)	M631 (kNm)	M632 (kNm)	M633 (kNm)	M634 (kNm)	M635 (kNm)	M636 (kNm)	M637 (kNm)	M638 (kNm)	M639 (kNm)	M640 (kNm)	M641 (kNm)	M642 (kNm)	M643 (kNm)	M644 (kNm)	M645 (kNm)	M646 (kNm)	M647 (kNm)	M648 (kNm)	M649 (kNm)	M650 (kNm)	M651 (kNm)	M652 (kNm)	M653 (kNm)	M654 (kNm)	M655 (kNm)	M656 (kNm)	M657 (kNm)	M658 (kNm)	M659 (kNm)	M660 (kNm)	M661 (kNm)	M662 (kNm)	M663 (kNm)	M664 (kNm)	M665 (kNm)	M666 (kNm)	M667 (kNm)	M668 (kNm)	M669 (kNm)	M670 (kNm)	M671 (kNm)	M672 (kNm)	M673 (kNm)	M674 (kNm)	M675 (kNm)	M676 (kNm)	M677 (kNm)	M678 (kNm)	M679 (kNm)	M680 (kNm)	M681 (kNm)	M682 (kNm)	M683 (kNm)	M684 (kNm)	M685 (kNm)	M686 (kNm)	M687 (kNm)	M688 (kNm)	M689 (kNm)	M690 (kNm)	M691 (kNm)	M692 (kNm)	M693 (kNm)	M694 (kNm)	M695 (kNm)	M696 (kNm)	M697 (kNm)	M698 (kNm)	M699 (kNm)	M700 (kNm)	M701 (kNm)	M702 (kNm)	M703 (kNm)	M704 (kNm)	M705 (kNm)	M706 (kNm)	M707 (kNm)	M708 (kNm)	M709 (kNm)	M710 (kNm)	M711 (kNm)	M712 (kNm)	M713 (kNm)	M714 (kNm)	M715 (kNm)	M716 (kNm)	M717 (kNm)	M718 (kNm)	M719 (kNm)	M720 (kNm)	M721 (kNm)	M722 (kNm)	M723 (kNm)	M724 (kNm)	M725 (kNm)	M726 (kNm)	M727 (kNm)	M728 (kNm)	M729 (kNm)	M730 (kNm)	M731 (kNm)	M732 (kNm)	M733 (kNm)	M734 (kNm)	M735 (kNm)	M736 (kNm)	M737 (kNm)	M738 (kNm)	M739 (kNm)	M740 (kNm)	M741 (kNm)	M742 (kNm)	M743 (kNm)	M744 (kNm)	M745 (kNm)	M746 (kNm)	M747 (kNm)	M748 (kNm)	M749 (kNm)	M750 (kNm)	M751 (kNm)	M752 (kNm)	M753 (kNm)	M754 (kNm)	M755 (kNm)	M756 (kNm)	M757 (kNm)	M758 (kNm)	M759 (kNm)	M760 (kNm)	M761 (kNm)	M762 (kNm)	M763 (kNm)	M764 (kNm)	M765 (kNm)	M766 (kNm)	M767 (kNm)	M768 (kNm)	M769 (kNm)	M770 (kNm)	M771 (kNm)	M772 (kNm)	M773 (kNm)	M774 (kNm)	M775 (kNm)	M776 (kNm)	M777 (kNm)	M778 (kNm)	M779 (kNm)	M780 (kNm)	M781 (kNm)	M782 (kNm)	M783 (kNm)	M784 (kNm)	M785 (kNm)	M786 (kNm)	M787 (kNm)	M788 (kNm)	M789 (kNm)	M790 (kNm)	M791 (kNm)	M792 (kNm)	M793 (kNm)	M794 (kNm)	M795 (kNm)	M796 (kNm)	M797 (kNm)	M798 (kNm)	M799 (kNm)	M800 (kNm)	M801 (kNm)	M802 (kNm)	M803 (kNm)	M804 (kNm)	M805 (kNm)	M806 (kNm)	M807 (kNm)	M808 (kNm)	M809 (kNm)	M810 (kNm)	M811 (kNm)	M812 (kNm)	M813 (kNm)	M814 (kNm)	M815 (kNm)	M816 (kNm)	M817 (kNm)	M818 (kNm)	M819 (kNm)	M820 (kNm)	M821 (kNm)	M822 (kNm)	M823 (kNm)	M824 (kNm)	M825 (kNm)	M826 (kNm)	M827 (kNm)	M828 (kNm)	M829 (kNm)	M830 (kNm)	M831 (kNm)	M832 (kNm)	M833 (kNm)	M834 (kNm)	M835 (kNm)	M836 (kNm)	M837 (kNm)	M838 (kNm)	M839 (kNm)	M840 (kNm)	M841 (kNm)	M842 (kNm)	M843 (kNm)	M844 (kNm)	M845 (kNm)	M846 (kNm)	M847 (kNm)	M848 (kNm)	M849 (kNm)	M850 (kNm)	M851 (kNm)	M852 (kNm)	M853 (kNm)	M854 (kNm)	M855 (kNm)	M856 (kNm)	M857 (kNm)	M858 (kNm)	M859 (kNm)	M860 (kNm)	M861 (kNm)	M862 (kNm)	M863 (kNm)	M864 (kNm)	M865 (kNm)	M866 (kNm)	M867 (kNm)	M868 (kNm)	M869 (kNm)	M870 (kNm)	M871 (kNm)	M872 (kNm)	M873 (kNm)	M874 (kNm)	M875 (kNm)	M876 (kNm)	M877 (kNm)	M878 (kNm)	M879 (kNm)	M880 (kNm)	M881 (kNm)	M882 (kNm)	M883 (kNm)	M884 (kNm)	M885 (kNm)	M886 (kNm)	M887 (kNm)	M888 (kNm)	M889 (kNm)	M890 (kNm)	M891 (kNm)	M892 (kNm)	M893 (kNm)	M894 (kNm)	M895 (kNm)	M896 (kNm)	M897 (kNm)	M898 (kNm)	M899 (kNm)	M900 (kNm)	M901 (kNm)	M902 (kNm)	M903 (kNm)	M904 (kNm)	M905 (kNm)	M906 (kNm)	M907 (kNm)	M908 (kNm)	M909 (kNm)	M910 (kNm)	M911 (kNm)	M912 (kNm)	M913 (kNm)	M914 (kNm)	M915 (kNm)	M916 (kNm)	M917 (kNm)	M918 (kNm)	M919 (kNm)	M920 (kNm)	M921 (kNm)	M922 (kNm)	M923 (kNm)	M924 (kNm)	M925 (kNm)	M926 (kNm)	M927 (kNm)	M928 (kNm)	M929 (kNm)	M930 (kNm)	M931 (kNm)	M932 (kNm)	M933 (kNm)	M934 (kNm)	M935 (kNm)	M936 (kNm)	M937 (kNm)	M938 (kNm)	M939 (kNm)	M940 (kNm)	M941 (kNm)	M942 (kNm)	M943 (kNm)	M944 (kNm)	M945 (kNm)	M946 (kNm)	M947 (kNm)	M948 (kNm)	M949 (kNm)	M950 (kNm)	M951 (kNm)	M952 (kNm)	M953 (kNm)	M954 (kNm)	M955 (kNm)	M956 (kNm)	M957 (kNm)	M958 (kNm)	M959 (kNm)	M960 (kNm)	M961 (kNm)	M962 (kNm)	M963 (kNm)	M964 (kNm)	M965 (kNm)	M966 (kNm)	M967 (kNm)	M968 (kNm)	M969 (kNm)	M970 (kNm)	M971 (kNm)	M972 (kNm)	M973 (kNm)	M974 (kNm)	M975 (kNm)	M976 (kNm)	M977 (kNm)	M978 (kNm)	M979 (kNm)	M980 (kNm)	M981 (kNm)	M982 (kNm)	M983 (kNm)	M984 (kNm)	M985 (kNm)	M986 (kNm)	M987 (kNm)	M988 (kNm)	M989 (kNm)	M990 (kNm)	M991 (kNm)	M992 (kNm)	M993 (kNm)	M994 (kNm)	M995 (kNm)	M996 (kNm)	M997 (kNm)	M998 (kNm)	M999 (kNm)	M1000 (kNm)	M1001 (kNm)	M1002 (kNm)	M1003 (kNm)	M1004 (kNm)	M1005 (kNm)	M1006 (kNm)	M1007 (kNm)	M1008 (kNm)	M1009 (kNm)	M1010 (kNm)	M1011 (kNm)	M1012 (kNm)	M1013 (kNm)	M1014 (kNm)	M1015 (kNm)	M1016 (kNm)	M1017 (kNm)	M1018 (kNm)	M1019 (kNm)	M1020 (kNm)	M1021 (kNm)	M1022 (kNm)	M1023 (kNm)	M1024 (kNm)	M1025 (kNm)	M1026 (kNm)	M1027 (kNm)	M1028 (kNm)	M1029 (kNm)	M1030 (kNm)	M1031 (kNm)	M1032 (kNm)	M1033 (kNm)	M1034 (kNm)	M1035 (kNm)	M1036 (kNm)	M1037 (kNm)	M1038 (kNm)	M1039 (kNm)	M1040 (kNm)	M1041 (kNm)	M1042 (kNm)	M1043 (kNm)	M1044 (kNm)	M1045 (kNm)	M1046 (kNm)	M1047 (kNm)	M1048 (kNm)	M1049 (kNm)	M1050 (kNm)	M1051 (kNm)	M1052 (kNm)	M1053 (kNm)	M1054 (kNm)	M1055 (kNm)	M1056 (kNm)	M1057 (kNm)	M1058 (kNm)	M1059 (kNm)	M1060 (kNm)	M1061 (kNm)	M1062 (kNm)	M1063 (kNm)	M1064 (kNm)	M1065 (kNm)	M1066 (kNm)	M1067 (kNm)	M1068 (kNm)	M1069 (kNm)	M1070 (kNm)	M1071 (kNm)	M1072 (kNm)	M1073 (kNm)	M1074 (kNm)	M1075 (kNm)	M1076 (kNm)	M1077 (kNm)	M1078 (kNm)	M1079 (kNm)	M1080 (kNm)	M1081 (kNm)	M1082 (kNm)	M1083 (kNm)	M1084 (kNm)	M1085 (kNm)	M1086 (kNm)	M1087 (kNm)	M1088 (kNm)	M1089 (kNm)	M1090 (kNm)	M1091 (kNm)	M1092 (kNm)	M1093 (kNm)	M1094 (kNm)	M1095 (kNm)	M1096 (kNm)	M1097 (kNm)	M1098 (kNm)	M1099 (kNm)	M1100 (kNm)	M1101 (kNm)	M1102 (kNm)	M1103 (kNm)	M1104 (kNm)	M1105 (kNm)	M1106 (kNm)	M1107 (kNm)	M1108 (kNm)	M1109 (kNm)	M1110 (kNm)	M1111 (kNm)	M1112 (kNm)	M1113 (kNm)	M1114 (kNm)	M1115 (kNm)	M1116 (kNm)	M1117 (kNm)	M1118 (kNm)	M1119 (kNm)	M1120 (kNm)	M1121 (kNm)	M1122 (kNm)	M
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COL A5 L1

General column design by PROKON. (GenCol Ver W2.2.06 - 01 Dec 2006)

Design code : CP65 - 1999

Input tables

General design parameters:

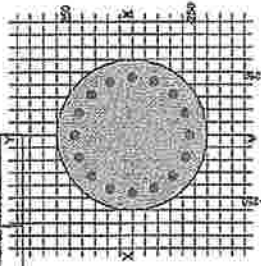
Code	X/Radius or Bar dia. (mm)	Y (mm) Angle (°)
a	300.000	
c	212.492	88.017
b	40.000	
b	162.635	162.635
b	40.000	
a	88.017	212.492
b	40.000	
a	-40.000	230.000
b	40.000	
a	-88.017	212.492
b	40.000	
a	-162.635	162.635
b	40.000	
a	-212.492	88.017
b	40.000	
a	-230.000	-0.000
b	40.000	
a	-212.492	-88.017
b	40.000	
a	-40.000	-230.000
b	40.000	
a	88.017	-212.492
b	40.000	
a	162.635	-162.635
b	40.000	
a	88.017	-212.492
b	40.000	
a	212.492	-88.017
b	40.000	
a	230.000	0.000
b	40.000	



Design loads:

Load case	Designation	P (kN)	Mx top (kNm)	My top (kNm)	Mx bot (kNm)	My bot (kNm)
1		6115	536			

Ultimate limit state design loads



General design parameters:

Given:
 L₀ = 4.500 m
 f_{cu} = 40 MPa
 f_y = 460 MPa
 A_c = 262637 mm²

Assumptions:

- (1) The general conditions of clause 3.8.1 are applicable.
- (2) The specified design axial loads include the self-weight of the column.
- (3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:
 (1) The column design charts are constructed.
 (2) The design axis and design ultimate moment are determined.
 (3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:
 At the top end: Condition 1 (fully fixed).
 At the bottom end: Condition 1 (fully fixed).
 The column is braced.
 $\beta = 0.75$

Effective column height:
 $l_e = \beta \cdot L_0 = 3.375 \text{ m}$

Check if the column is slender:
 $l_e/h = 6.5 < 15$
 $\therefore$ The column is short.

Initial moments:

The initial end moments about the X-X axis:
 M₁ = Smaller initial end moment = 0.0 kNm
 M₂ = Larger initial end moment = 838.0 kNm
 The initial moment near mid-height of the column :
 $\therefore M_l = 0.4M_1 + 0.6M_2 \geq 0.4M_2 = 502.8 \text{ kNm}$

The initial end moments about the Y-Y axis:
 M₁ = Smaller initial end moment = 0.0 kNm
 M₂ = Larger initial end moment = 0.0 kNm

The initial moment near mid-height of the column :
 $\therefore M_l = 0.4M_1 + 0.6M_2 \geq 0.4M_2 = 0.0 \text{ kNm}$

Table 3.21

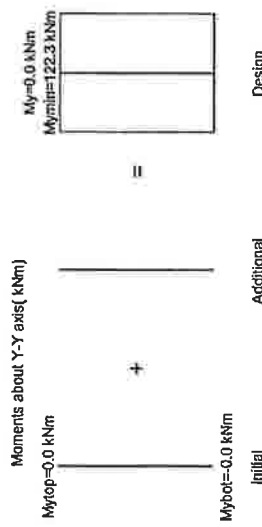
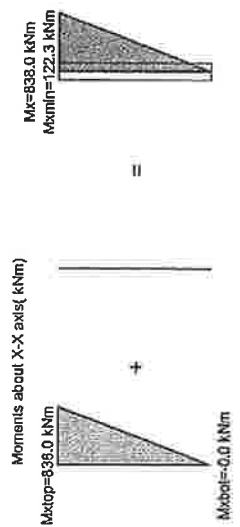
3.8.1.3

3.8.3.2

3.8.3.2

Design ultimate load and moment:

Design axial load:
 $P_u = 6115.0 \text{ kN}$



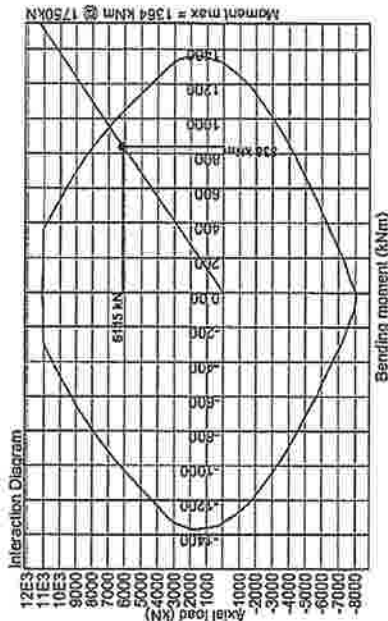
Check for minimum eccentricity:
 Check that the eccentricity exceeds the minimum in the plane of bending:
 Use $e_{\text{min}} = 20 \text{ mm}$
 $\therefore M_{\text{min}} = 122.3 \text{ kNm}$ about the X-X axis.

3.8.2.4

Design of column section for ULS:

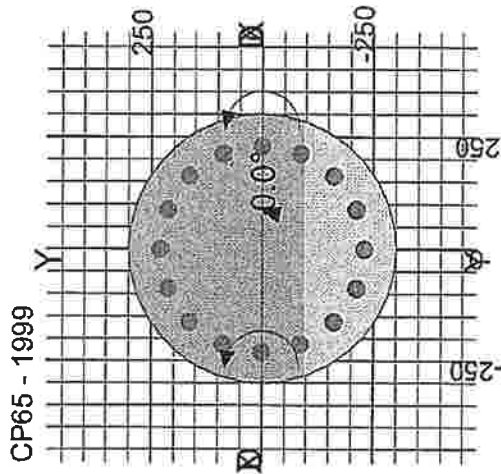
The column is checked for applied moment about the design axis.
 Through inspection: the critical section lies at the top end of the column.
 The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
 The safety factor for the critical load case 1 is 1.15

For bending about the design axis:



Moment distribution along the height of the column for bending about the design axis:
 At the top, $M_x = 838.0 \text{ kNm}$
 Near mid-height, $M_x = 502.8 \text{ kNm}$
 At the bottom, $M_x = 122.3 \text{ kNm}$

Stresses at the top end of the column for the critical load case 1



Summary of design calculations:

Design results for all load cases:

Load case	Axis	N (kN)	M1 (kNm)	M2 (kNm)	Mi (kNm)	Madd (kNm)	Design	M (kNm)	M _r (kNm)	Safety factor
Load case 1	X-X	6115.0	0.0	838.0	502.8	0.0	Top	838.0	838.0	1.146
	Y-Y		0.0	0.0	0.0	0.0		0.0		

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COL A5 L1

General column design by PROKON. (GenCol Ver W2.2.08 - 01 Dec 2008)

Design code : CP65 - 1999

C13

Input tables

General design parameters:

Code	X/Radius or Bar dia. (mm)	Y (mm) Angle (°)
a	300.000	
b	212.492	88.017
c	40.000	
d	162.635	162.635
e	40.000	
f	88.017	212.492
g	40.000	
h	-0.000	230.000
i	40.000	
j	-88.017	212.492
k	40.000	
l	-162.635	162.635
m	40.000	
n	-212.492	88.017
o	40.000	
p	-162.635	-162.635
q	40.000	
r	-88.017	-212.492
s	40.000	
t	0.000	-230.000
u	40.000	
v	88.017	-212.492
w	40.000	
x	162.635	-162.635
y	40.000	
z	212.492	-88.017
aa	40.000	
ab	230.000	0.000
ac	40.000	

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Design loads:

Load case	Designation	P (kN)	Mx top (kNm)	My top (kNm)	Mx bot (kNm)	My bot (kNm)
1		6115	838			

Ultimate limit state design loads

General design parameters:

Given:

Lo = 4.300 m

fcd = 40 MPa

fy = 460 MPa

Ac = 262637 mm²

Assumptions:

- (1) The general conditions of clause 3.8.1 are applicable.
- (2) The specified design axial loads include the self-weight of the column.
- (3) The design axial loads are taken constant over the height of the column.

Design approach:

The column is designed using the following procedure:

- (1) The column design charts are constructed.
- (2) The design axis and design ultimate moment are determined.
- (3) The design axial force and moment capacity is checked on the relevant design chart.

Check column slenderness:

End fixity and bracing for bending about the Design axis:

At the top end: Condition 1 (fully fixed).

At the bottom end: Condition 1 (fully fixed).

The column is unbraced.

$\beta = 1.20$

Effective column height:

le = β · Lo = 5.160 m

Check if the column is slender:

le/h = 10.0 < 10

∴ The column is short.

Initial moments:

The initial end moments about the X-X axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 838.0 kNm

The initial moment near mid-height of the column:

∴ Ml = 0.4M1 + 0.6M2 ≥ 0.4M2 = 502.8 kNm

The initial end moments about the Y-Y axis:

M1 = Smaller initial end moment = 0.0 kNm

M2 = Larger initial end moment = 0.0 kNm

The initial moment near mid-height of the column:

∴ Ml = 0.4M1 + 0.6M2 ≥ 0.4M2 = 0.0 kNm

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Design ultimate load and moment:
 Design axial load:
 $P_u = 6115.0 \text{ kN}$

Moments about X-X axis (kNm)

Moments about Y-Y axis (kNm)

Check for minimum eccentricity:
 Check that the eccentricity exceeds the minimum in the plane of bending:
 Use $e_{min} = 20 \text{ mm}$
 $\therefore M_{min} = 122.3 \text{ kNm}$ about the X-X axis.

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Design of column section for ULS:

The column is checked for applied moment about the design axis.
 Through inspection: the critical section lies at the top end of the column.
 The design axis for the critical load case 1 lies at an angle of 0.00° to the X-axis
 The safety factor for the critical load case 1 is 1.15

For bending about the design axis:

Moment distribution along the height of the column for bending about the design axis:
 At the top, $M_x = 838.0 \text{ kNm}$
 Near mid-height, $M_x = 502.8 \text{ kNm}$
 At the bottom, $M_x = 122.3 \text{ kNm}$

